ENGLISH

Chapter 1

# Sleep

Sleep is the subject of the first chapter of this book because sleep profoundly impacts every aspect of our physiology and psychology. Put simply, sleep is the foundation of our mental health, our physical health, and our performance. We win by sleeping. I came to that realization pretty early but only because I struggled with sleep. As a kid I slept like a log—always a solid eight to nine hours, rarely waking up in the middle of the night even once. As a teen I would stay out late and sleep in on the weekends, as most of us did. I never thought about my sleep. When I reached my junior year in high school, however, everything changed. I would lie down, fall asleep right away, but then wake up three to four hours later, wide awake. Then I would toss and turn. Eventually I would fall back asleep but then needed to sleep really late into the morning to feel rested. School started at 8:00 a.m., so that wasn't an option most days. That had me falling asleep in class, feeling wide awake in the evening, and then the whole cycle would repeat.

Toward the end of high school when I started getting into fitness, I learned that we recover from exercise during sleep. That was the early 1990s and the public was not yet aware of the importance of sleep. There was some science on it—in fact some of the best work on the role of sleep for health and cognition was coming from the Stanford Sleep Lab just a few miles up the road from where I lived. But it was from a prominent trainer in the world of fitness that I first learned to take sleep seriously. He gave me one mostly good suggestion for sleep, and one terrible one. The mostly good suggestion was to read the work of Allan Hobson—a sleep researcher from Harvard—and I eventually did, although I discovered Allan's focus was mainly on the importance of dreams, which felt like a bit of a tease for someone with sleep issues. The terrible advice he gave me? He told me if I woke up in the middle of the night and couldn't fall back asleep that I should drink a half mug of white wine. I was seventeen! I tried it anyway, and it didn't work for me. Thankfully. We now know alcohol greatly reduces the quality and restorative properties of sleep, even if it does help some people fall asleep.

Eventually I figured out how to master the art of getting a great night's sleep, but it took time. Years, in fact. The science progressed too, of course, and we now know a lot about both the importance of sleep and ways to remedy pretty much any sleep issue. I wish I had the information you are about to read back in high school, and college, and graduate school, and my postdoc. The reality is, the busier we are, the more important sleep is to our health and performance. But no matter how busy or calm our life is, a great night's sleep is crucial to our mind and body. If

you are like the majority of people, and you struggle to get a great night's sleep, you want to remedy that as soon as possible. But even if you are one of those rare individuals who sleeps deeply every night and wakes up feeling fully refreshed every day, there are still ways to improve your sleep and thus, your quality of life.

joyful experiences, like a great party. Do your best to make sleep a top priority so that the rest of your life can be that much better.

Perhaps you have heard some of this before. Most people know a bit about the benefits and importance of sleep. As I mentioned above, that is a fairly recent development. It was about a decade ago when Matthew Walker, who formerly directed the Center for Human Sleep Science at the University of California, Berkeley, and is currently at the University of Texas in Dallas, wrote the now famous book, Why We Sleep. At that time, the "I'll sleep when I'm dead" or the "I'll catch up on sleep this weekend" mentalities prevailed. Thanks in large part to Matt's book and other public education efforts, people now recognize they must sleep, yet the majority of people still don't sleep well and far too many people sleep poorly, with nearly half of adults reporting their sleep satisfaction as an "F"—meaning the lowest possible grade.

It is true that we all have slightly different sleep needs. Some of us consider ourselves morning larks. We prefer to rise early. Others are night owls who prefer to stay up late. These are what sleep researchers call chronotypes: genetically determined features of the chemicals, cells, and circuits that regulate when we prefer to go to sleep and wake up. You can blame your parents for your bedtime (and wake time) preference. Separate from your chronotype is the total amount of sleep you need. Some of us need six hours to feel fully rested, others need eight, and some people feel great with just five. I should point out, however, that the feel-great-on-five folks are exceedingly rare. Most people need seven to eight.

But as mentioned above, according to recent reports from the National Sleep Foundation and the Centers for Disease Control and Prevention (CDC), at least a third of Americans do not get nearly enough sleep—and that statistic is likely to be a vast underestimate given the enormous increase in time spent on social media and online nowadays. In fact, that figure skyrockets to more than 75 percent among high school students. That's right, approximately three-fourths of teens are not getting enough sleep, which is extremely concerning given that our adolescent and teen years are when much of our brain wiring related to mood and behavior regulation, as well as attention and learning, gets solidified. From birth

until about age twenty-five is an especially sensitive and time-limited phase, or what is called a critical period. And the thing about critical periods is eventually they close: While our brains remain plastic our entire lives, the amount of neuroplasticity possible in those first twenty-five years far exceeds what is possible afterward, so teens can't simply make up for their lost brain wiring later on. We have no idea what the long-term consequences of this kind of chronic sleep deprivation are, but we do know that far too many young people are missing out on sleep at exactly the time when they need it most.

Compulsive overuse of phones and other screen-based technologies at night is a major factor for the poor sleep of adolescents and teens, but also for the poor sleep habits of adults. A central feature of life in the twenty-first century is to always be connected online and by text. So much so that by the classic definition of shift work, many—perhaps even most people—worldwide, now meet the definition of a shift worker, that is, awake for two to three hours between 10:00 p.m. and 5:00 a.m. Shift work is not just about staying up late but also involves reduced amounts of exposure to sunlight during the day, lack of sufficient darkness at night, and a nocturnally-shifted eating schedule. That has well-established negative health effects, including poor gut health, mental health deficits, and even decreased lifespan. So this is no trivial matter.

Smartphone technologies are not to blame per se, but when we use those technologies certainly is. We can scroll 24-7 and unless we resist, the light from our phone, tablet, or computer screen impairs our sleep. Not to mention that the content we absorb on those screens often affects our sleep, too, because of the elevated levels of arousal—aka stress—that content creates. Even if being on our phones or tablets in the hours before sleep doesn't stress us out, it can disrupt our sleep. We rarely hear how texting alters our sleep patterns but I would argue that for many people, it is at least as pernicious as social media in this regard. To be clear, texting is a wonderful tool for communication; it allows us to connect quickly. And social media can be a good source for learning, social connection, and entertainment. The downside, however, is that we can have conversations and observe events at any time of day or night. If we wake up in the middle of the night, we know we should roll over and go back to sleep, but many of us instead start reading and sending texts, or scrolling the internet or browsing social media.

▸ View Sunlight as Soon as Possible After Waking

## WHAT TO DO

• Get morning sunlight in your eyes as soon as possible after waking. Go outside and look in the direction of the sun for at least five to ten minutes each morning. Your goal should be to get sunlight in your eyes within the first thirty to sixty minutes of waking up.

• A few things to keep in mind. If you wake up before the sun rises, you'll obviously have to wait. If you can view the sun as it comes up across the horizon, great, but that's not vital for this protocol to work. Morning sunlight, as we will review, differs in quality from sunlight at other times of the day. It is best viewed by facing the sun without any physical barriers, like awnings or buildings, obscuring your view. Don't wear a brimmed hat, as it blocks some of the light that would otherwise reach your eyes from overhead. Wearing eyeglasses or contacts is fine but, alas, this protocol won't work nearly as well through a window or windshield, as that will filter out

## HOW IT WORKS

up a robust network that drives many of the biological processes in your brain and body. I first learned about the importance of viewing morning sunlight from one of the world's preeminent sleep researchers, Seiji Nishino, MD, PhD. I was working in his lab at Stanford during a summer home from college, and I noticed he would always stop to face the sun for a few minutes before going into the lab for the day. Even if he arrived at the lab before sunrise (not uncommon for researchers in a sleep lab!), he would make sure to step outside shortly after sunrise and face the sun for a few minutes before returning to work. That was way back in 1996—when our understanding of the mechanisms of human circadian biology was rudimentary compared to now. Seiji explained that viewing morning sunlight felt good, that it gave him more energy for the day, and that it improved his sleep. I started joining him and joked that we resembled the many ground squirrels around us up on their haunches doing the very same thing. Little did I know that the exact same morning sunlight-sensing cells, circuits, and mechanisms were at play in us and those squirrels and, in fact, in every mammal where they have been examined. I quickly noticed that on days I viewed morning sunlight, I also felt good, had more energy for the day, and slept better at night, so I kept with it.

You should experience those benefits, too, because of the way morning sunlight—whether on a clear or cloudy day—kickstarts a powerful set of hormonal and neural mechanisms that positively shift the state of your

entire brain and body for many hours afterward. It can work the first time and every time, and once you understand a bit about those mechanisms, it will allow you to easily adjust this protocol to your schedule and circumstances to derive the maximum benefits with minimal time investment.

## Your Circadian Clock

Located a few centimeters above the roof of your mouth is a small cluster of neurons called the suprachiasmatic nucleus, or SCN. The SCN is your master circadian clock. Its job is to coordinate the activities of all the cells and organs in your body into a coherent twenty-four-hour rhythm so that certain cells and organs in your body are more active, and other cells and organs are less active, depending on the phase of that twenty-four-hour rhythm. The SCN keeps your body and brain on schedule.

The first documented discovery of a circadian clock came not from studies of humans or even of rodents, but of plants. In 1729, a French astronomer named Jean-Jacques d'Ortous de Mairan noticed that a mimosa plant kept in a totally dark room continued to rhythmically fold and unfold its leaves every twenty-four hours, just as if the sun was shining through the window. It was a mystery: If the plant was not unfurling to absorb sun or other light, then what was guiding its movements? About two hundred years later, German biologist Erwin Bünning tracked the movements of bean plant leaves and reported that they followed a cycle lasting 24.4 hours. Observations such as these confirmed that plants had internal, or what scientists call "endogenous," clocks that followed what later became known as... a circadian rhythm, named for the Latin circa diem, or "around a day."

And as decades of research would eventually confirm, the vast majority of organisms on Earth—not just plants but also insects, rodents, reptiles, and all mammals, including humans—have an endogenous timekeeper that keeps a twenty-four-hour circadian rhythm, even in the absence of any external cues about the status of the outside world. Our core body temperature, the release of certain hormones, and most notably our sleep-wake cycle, all follow a twenty-four-hour circadian rhythm. Even when scientists extracted individual cells from the body, the cells continued to follow this rhythm. And which genes they expressed and when, the proteins downstream of those genes, the cells' enzymatic activities—indeed not just some things but nearly everything about cellular function—appeared to vary, not randomly, but rather systematically every twenty-four hours. That was remarkable.

But what is equally remarkable is that all cells in our body can coordinate their individual rhythms in a manner that is timed perfectly with one another—like a well-practiced symphony, except this symphony involves not a few hundred or even a thousand participants, but trillions. And just like a symphony, it is not as if every cell simply needs to start and end its activity at the same time. Some cells need to be active at the same time, or in-phase with one another; some need to be active at opposite times, or in anti-phase, and still others repeat their cycles of activity faster, then slower, then faster, again with a unique signature and yet, still repeating once every twenty-four hours. And the start and

sealed bunker, living life as usual but without any windows or other ways to determine the passage of time. They found that subjects generally spent about one-third of each twenty-four-hour period sleeping and two-thirds of it awake, just as most humans living aboveground do. However, the wake-up and to-sleep times of the people in the bunker drifted—they woke up slightly later and went to sleep slightly later each day. Thus, the bunker experiment confirmed the existence of a circadian rhythm, but one that requires some external cue to anchor it.

What remained mysterious was the biological basis of the circadian clock—was it a bloodborne factor? In the liver? In the brain? In the pineal gland? Theories abounded but none of them held up to experimentation. And another major question remained: In the environment outside the bunker, which external cues were responsible for setting that clock to a specific start and finish time each day?

Subsequent discoveries, first in rodents and later in humans, revealed the existence of a small cluster—or what is called a "nucleus"—of neurons located a few centimeters above our mouths that had a key part to play. It was later named the suprachiasmatic nucleus (SCN) for "above the chiasm"—the chiasm being the location at the base of the brain where the nerve fibers from both eyes partially cross. As I mentioned earlier, the SCN is the master circadian clock: the structure that coordinates the activities of all our cells and organs. In other words, the SCN is the orchestra conductor that makes sure each cell's twenty-four-hour rhythm starts on time. In 1972, Irving Zucker's lab at UC Berkeley and Robert Moore's lab at the University of Chicago independently discovered that lesions to the SCN render organisms' circadian pattern arrhythmic and disorganized. Naturally occurring SCN lesions caused by tumors eventually showed the SCN is necessary for sleep-wake, hunger, and other bodily rhythms to obey circadian cycles in humans as well. Moreover, animal studies showed that transplanting even just a small number of healthy SCN neurons into an

arrhythmic brain restored a coherent twenty-four-hour rhythm, proving the SCN is both necessary and sufficient for endogenous circadian rhythms.

If ever there was a brain structure whose function is beautifully organized to help ensure our survival as a species, it is the SCN. Even the SCN's seeming imperfection turns out to be perfect after all. Stay with me here. For example, the SCN generates a period—meaning a cycle length—that is close to but not quite twenty-four hours. The SCN's period tends to be a bit longer than that, up to 24.5 hours. By virtue of this, it can shift the start of the circadian rhythm forward or backward according to seasonal changes in day length or other survival-based needs, without making a mess and setting our cells on different schedules. In other words, allowing a little room in its period duration affords us more flexibility and overall precision, not less, which is great because as we all know life is rarely the same from one day to the next. What's more, it is also what allows for phase shifting, or the ability to change the start time of the daily cycle. If you're wondering why this doesn't have us waking up later and going to sleep later each day, it's because viewing morning light and evening light nudges your clock back and forth perfectly, to keep it regular. Later I'll explain how you can leverage that to overcome jet lag or any new schedule fast, but for now, all you need to know is that the not-exactly-twenty-four-hour rhythm is a feature, not a flaw.

As you probably already realize from the call to action that is Sleep Protocol 1, our SCN cares about whether it is light or dark out. And of course light-dark cycles are set by the spin of the earth. But how is it that our SCN—a tiny cluster of neurons embedded deep in the darkness of our brains—"knows" about the spin of the earth? Let's consider how to answer that question with insights from planetary science and cosmology. We're all on this big sphere in space, spinning around at about one thousand miles per hour, causing us to face the sun, then away from the sun, then back toward the sun every twenty-four hours. Because the earth is round, the exact duration of sunlight we can experience each day depends, of course, on where we happen to be on earth. The variability is large, ranging from rather equal light-dark splits at the equator to nearly full days and/or nights of darkness or sunlight exposure at the poles. But no matter where we are on earth, inside each of our brains, our SCN steadily churns out that slightly longer than twenty-four-hour rhythm of coordinated cellular activity across our whole body. Ultimately it is our SCN that tells us when to sleep and when to wake. We are a diurnal, not a nocturnal, species. We can stay up late and sleep all day if we choose, but even for

optic nerves convey electrical signals about the presence or absence of sunlight, objects, faces, moving vehicles, and on and on to different brain areas, including—you guessed it—the SCN. The connections between your eyes and your brain are truly a spectacular product of biological engineering. First, they solve the problem of how to inform image-forming regions of your brain about "what is out there" in the world, so you can see faces, moving objects, and colors. They also solve the problem of letting your SCN know "when it is out there" in the world, by the presence of sunlight.

The neurons that convey the presence of sunlight to the SCN are not of the image-forming kind; they don't respond to the shape of the sun, but instead to its brightness and to the specific wavelengths of sunlight that reach your eyes when the sun is low in the sky. We refer to these specific wavelengths as low solar angle sunlight. Have you ever noticed that low solar angle sunlight appears more yellow than midday sun? Or even orange? Check out a sunrise or sunset and compare the color of sunlight to the color of sunlight at noon. Of course, sunlight doesn't change according to where it is in the sky, but what portions of sunlight are visible to us does. Sunlight is always full spectrum: It contains all wavelengths of light, red-orange-yellow-green-blue-indigo-violet, or ROYGBIV, just as they taught us in elementary school. The R(ed), O(range), and Y(ellow) are longer wavelengths of light, the G(reen) is medium wavelength, and the B(lue), I(ndigo), and V(iolet) are shorter wavelengths. And of course, the sun emits UV and far-red, or infrared wavelengths, too, but we can't see those (although many other animals and insects can). When the sun is high in the sky, it appears white, because all the wavelengths of light arrive at our eyes at roughly the same intensities. When the sun is low in the sky, we see the sun as more yellow, orange, or red, because molecules in the atmosphere preferentially redirect the shorter blue, indigo, and violet wavelengths of light—a process called Rayleigh scattering. (We'll discuss the properties of light in depth in chapter 5.)

All mammals, us humans included, have evolved a special system of light detection that responds specifically to the presence and contrast of short and long wavelengths of bright light, which is exactly what reaches your eyes if you view sunlight early or late in the day, while the sun is fairly low in the sky. In other words, we all have a sunrise and sunset detecting system in our eyes that communicates the time of day to our SCN.

As I mentioned earlier, this timekeeping eye-to-brain circuit is also a fairly recent discovery. And by recent, I mean recent: In the late 1990s, there still wasn't general agreement among circadian biologists whether

light—and not some other cue like social interactions—was the dominant force entraining our SCN. Then, in 1998, neuroscientist Ignacio "Iggy" Provencio made a key discovery. Iggy found a molecule in the skin of an African frog species that caused its skin to darken when exposed to sunlight was also present in the eyes of mice. The molecule was an opsin, which is a kind of protein that absorbs light, and when it does, it changes its conformation to set off a signaling cascade. Since the same opsin was known to cause darkening of dermal melanophores in frog skin, Provencio called the mouse retinal opsin melanopsin. Other labs confirmed melanopsin-containing cells are abundant in the human retina, too, mostly in the lower portion that views the upper visual field, which, you might have guessed, corresponds to the sky.

So we have a frog-skin-like set of cells in our eyes? Not quite. To explain what makes melanopsin-containing retinal cells so special, first a bit about photoreceptors. Photoreceptors are neurons that convert light into electrical signals the rest of the retina and brain can interpret. But the retinal cells that contain melanopsin are not like other photoreceptors. The melanopsin-containing retinal neurons Iggy discovered turned out to be both photoreceptive and retinal ganglion cells—the neurons that connect the eyes to the brain and the very cells whose wirelike extensions, or axons, make up the optic nerves. Soon after, a group of neuroscientists, David Berson and Felice Dunn at Brown University and Samer Hattar and King-Wai Yau at Johns Hopkins University, along with several of their students and postdocs, confirmed that melanopsin RGCs send their axons to the SCN and that they fire electric signals, thereby releasing neurotransmitters, when light is shone on them. Remarkably, they did this when they were isolated from all other retinal cells. As long as the light shone was very bright and long lasting, they fired electrical signals. These cells were therefore named intrinsically photosensitive RGCs (ipRGCs) to differentiate them from the other 99 percent of RGCs that can't

Doing so activates the ipRGC-to-SCN circadian entrainment circuit and initiates dozens of beneficial processes that get and keep your brain and body going throughout the day. And toward the evening, it puts you in the best possible position to get a great night's sleep.

In fact, we have copious evidence that this is foundational to both our daytime well-being and nighttime sleep. As one small example, a group of architects and neurologists came together in 2014 to investigate how windowless offices affected the people who worked in them. They asked twenty-seven office workers who had either no windows, or work areas situated far from windows, and twenty-two people working in offices with

windows, about their well-being and sleep quality. The questionnaire scores were decidedly poorer among the windowless workers. They reported more sleep disturbances and an overall unsatisfactory quality of sleep.

In the early 2000s, a group of chronobiologists (scientists who study biological timing) at the Institute for Medical Psychology, in Munich, Germany, surveyed five hundred people about their sleep times, light exposure, and chronotype—as in, morning person, night owl, et cetera (you'll find out how to determine yours in Sleep Protocol 4). They found that the amount of light exposure during the day correlated with what time people went to bed: More exposure during the day meant earlier bedtimes. This link can be partly explained by external factors, like employment: People whose livelihoods depend on being outdoors are more likely to be early-to-bed types, especially if their work requires them to rise early or is physically demanding. But that doesn't explain the connection entirely. It turns out that it's also due, in part, to the fact that exposure to daylight has a regulating effect on our physiology—causing elevated levels of alertness by a host of different pathways, including hormonal and neural pathways. As one review put it, "In sum, research seems to agree that daylight (at high intensities) is beneficial for sleep." There are also numerous studies showing that bright-light exposure early in the day—at the intensities that one would get from being outdoors in the morning—is beneficial for everything from mood to hormone health, daytime energy levels, and on and on.

But before you say, "Okay, got it!," there's just a bit more to understand about the underlying mechanisms in this protocol, and that's the relationship between sleep and wakefulness. And it will help you get the most out of this protocol, regardless of where you live and the time of year.

Our daytime levels of alertness reflect the quality and duration of our nighttime sleep. That's a giant "duh." We all know that if we sleep poorly, we drag both mentally and physically the next day. We also all know that if we sleep great, we're focused and motivated to do all kinds of things that require energy... and that we have the energy to do them. But what most people do not know is that the relationship between sleep and wakefulness works in the other direction too. When we are alert during the morning and daytime, we are more likely to expend mental and physical energy that in turn helps us sleep at night. Even those of us who sleep well typically find that increasing our daytime physical activity—especially in the hours between sunrise and 2:00 p.m.—leads to significantly better sleep, which in turn leads to more energy the next day, which improves sleep, and so on.

Again, you're probably thinking, "This is common knowledge. Do we really need science to explain this?" Here is the lesser-known fact: Morning sunlight viewing is the best way to increase our morning and daytime energy levels. Not food (although fine to eat early in the day), not caffeine (also fine). It's morning sunlight.

Morning sunlight increases your energy levels through at least three different mechanisms. First, it suppresses any residual melatonin you may have circulating in your brain (Sleep Protocol 3 includes an extensive look at melatonin, a hormone that helps promote sleepiness). As discussed above, when the ipRGCs are activated by sunlight viewing, they send electrical signals to the SCN. The SCN, in turn, stimulates a collection of neurons in the nearby paraventricular nucleus, or PVN. The PVN then sets off a cascade of signals that go through the spinal cord and back up to the pineal gland in the brain, shutting off melatonin production. Voila! You've suppressed the primary hormone that promotes sleepiness by viewing morning sunlight.

Second, bright light reaching your eyes helps suppress the action of adenosine, a chemical that also leads to feelings of sleepiness. Sunlight does this indirectly via its effects on other systems in the brain and body, but nonetheless, it has a potent adenosine-reducing effect. (We'll take a deep dive into adenosine in Sleep Protocol 7). After a full night's sleep, adenosine levels tend to be low, but if you happen to not get enough sleep (as so many of us do), morning light helps reduce any residual adenosine. That both increases your alertness and prevents the residual adenosine from delaying your circadian clock, which would otherwise bias you toward waking up later the next day, then going to sleep later, then waking up even later the following day, and so on (what chronobiologists call a phase delay). Such circadian "drift" brought on by mild sleep deprivation is very common. It is best remedied by viewing morning sunlight to reset your circadian rhythm.

Third, we need to talk about cortisol. Because it's often discussed in the context of chronic stress, which is bad, most people think cortisol is "bad" too. But it's not bad. Cortisol is essential to health and your levels of energy—you just don't want your cortisol levels to be elevated at the wrong times. Your SCN is located in your hypothalamus (H), which is part of a biological circuit that also includes your pituitary gland (P), a small, stalklike protrusion at the base of your brain that releases hormones into your bloodstream. The circuit also includes your adrenal glands (A), which reside atop your kidneys. This circuit is known as the HPA axis. The HPA axis is one of the main systems that responds to stress (which is why you'll

hear more about it in chapter 3). Here's how it works: It starts with signals from the hypothalamus, which then stimulate the release of specific hormones from the pituitary, which in turn stimulate adrenaline and cortisol release from your adrenals.

and late afternoon. Cortisol levels generally stay low and continue to drop in the evening and at nighttime (assuming it doesn't spike from stress or intense exercise—more on that below), and then it falls to its nadir in those first four to five hours of sleep before starting to rise again toward morning. We return to this circadian cortisol curve in chapter 3, and the role it plays in promoting health in women and men of all ages. For now, the key thing to know is that for cortisol to support our health, it needs to peak in the first hour or so after waking; it then needs to drop precipitously in the evening and night and continue falling to reach its nadir in those initial four to five hours of sleep. Viewing morning sunlight doesn't directly cause the morning peak in cortisol, but it can dramatically amplify its rise by as much as 50 percent. This also helps ensure that this peak comes early in the day and isn't shifted to the afternoon. Late-day elevations in cortisol have negative consequences over time such as sleeplessness, anxiety, and metabolic disease.

A quick note: A morning workout, morning caffeine, and breakfast (provided it isn't a huge and/or carbohydrate-laden one) will all promote wakefulness, too, and thereby contribute to a well-aligned circadian rhythm. But morning sunlight is the most potent lever for increasing morning cortisol and thus energy during your day, and for improving your sleep at night. Don't underestimate its power. Keep your circadian rhythm attuned to the sun, and all of these mechanisms will work as they should.

## FREQUENTLY ASKED QUESTIONS

### It's often dark when I wake up. What do I do if I have no access to morning sunlight but I want to start my day?

If you wake up before sunrise, and you want to start your day, I suggest turning on as many bright indoor lights in your home—or hotel room, or wherever you may be—as possible. Sunlight cannot be replaced by standard indoor lighting, but turning on indoor lighting first thing won't hurt. Then get outside to view morning sunlight once the sun is up.

If you wake before sunrise every day, or if you live in a part of the world where it is very dark in winter, you may want to invest in a 10,000 lux light to help ensure you get enough light in the morning (lux is a measure of brightness). Most cost about $100, and some are portable for travel. I often

wake up before sunrise, and because the ipRGCs can be activated by very bright artificial lights if those bright lights include sufficient short wavelengths, I keep a 10,000 lux lamp in my kitchen and bathroom. It lacks long wavelength light, so I also have a red incandescent light next to it and I turn that on too. I use the red incandescent light from SaunaSpace (no affiliation). I turn them both on first thing in the morning and then I get outside to view morning sunlight once the sun is up. I am well aware these lights are not a replacement for sunlight, but they help bridge the gap until the sun is up. Once the sun is up, you can be sure I go outside to view it! And I will sometimes keep the 10,000 lux lamp and a red incandescent light on and facing me for an hour or so once I'm back working at my desk or getting ready for the day. Chapter 5 further explains the logic and benefits of this extended morning light exposure. There I also mention options that can be of real help for sleep as well as other aspects of health too, such as metabolic health.

But remember, morning sunlight is the best stimulus for your ipRGCs, SCN, and circadian rhythm, and it costs nothing! Figure out how to view it each morning. Again, you don't need to see it cross the horizon as it rises. If you do, great, but if you wake up after sunrise, that's fine. Do your best to view sunlight within the first thirty to sixty minutes after waking up. Bring your coffee or tea outside with you. Bring your spouse or kids if they're awake. Bring your dog if you have one. Remember, dogs, cats, cows, and other animals have ipRGCs and an SCN, too, and they benefit from morning sunlight in many of the same ways and via the same mechanisms as we do.

**What if it's cloudy outside?**

To activate your ipRGCs fastest, go outside and face east toward the sun on a nice, clear morning with no clouds in the sky. Obviously, this isn't possible for everyone every day. Again, fortunately, sparse clouds do not interfere with the interaction between sunlight and our circadian clock. However, if the sky is densely overcast and gray, a 10,000 lux light can be a real help, but still aim for getting more sunlight in your eyes to compensate for the lower intensity of sunlight and reduced yellow-blue contrast exposure you're likely to get on very overcast days. In fact, it's especially important that you get sunlight in your eyes on overcast days. This is for three reasons: (1) dense clouds reduce the number of photons reaching your eyes in a given amount of time; (2) they filter out much of the longer (yellow, orange, and red) wavelengths of light, and (3) you are likely to be indoors more on cloudy days, which can cause the SCN to fall out of sync with the solar cycle, which leads to later nights, more artificial

light exposure at night, more circadian disruption, and so on.

Aim to extend the sunny, clear day recommendation of a minimum of five to ten minutes, to at least fifteen minutes on cloudy days, and thirty minutes (minimum) on densely overcast days. That may seem like a long time, but remember, you don't need to stand in one place gazing at the sky. Take a walk. Make a call. Listen to a book or podcast or talk to a friend, partner, or other family member. And even if you can't be outside for the full time recommended above, some sunlight hitting your eyes in the morning is better than none. If you miss a day completely due to travel or sickness or other reasons, don't despair—your energy levels that day may be lower because you didn't get the cortisol spike bright morning light causes, but your circadian rhythm is the reflection of your previous three to four days of sunlight viewing activity.

**Do I need to worry about sunlight damaging my eyes?**

Excessive UV light exposure can cause cataracts and, as we have all heard, skin damage, accelerated skin aging, and some forms of skin cancer, topics I cover in more detail later in the book. But viewing morning sunlight is much safer than viewing overhead midday sunlight—because that Rayleigh scattering of short

People relentlessly ask me this and my response is equally persistent: No, it's not the same. It's far less effective, though better than nothing. But if there is any way to get outside, do. Windows and windshields, especially nowadays, filter the intensities and wavelengths you want for this protocol to work. "How come viewing morning sunlight through eyeglasses or contacts is fine but not windows or windshields?" you may wonder. Because windows do something called spectral filtering: They reduce the amount of light—especially short wavelength light—that would otherwise reach your eyes. Corrective lenses don't; they let far more light through. That's their job. Well, that and to correct any optical errors in the structure of your eyeballs so you can see things clearly. Windows and windshields do neither.

▸ View Sunlight in the Late Afternoon

## WHAT TO DO

• Get sunlight in your eyes in the late afternoon or early evening. Ideally, go outside and spend at least ten to fifteen minutes there before the sun sets. If at all possible, face or walk west toward the sun. You do not need to see the sun drop below the horizon. That's a wonderful thing

if you can—who doesn't love a sunset? Alas, most of us don't live or work in the right place to do so. Luckily, just going outside and getting some ambient light in your eyes in the late afternoon or early evening is beneficial. Seeing the sunset is a bonus.

• A few things to keep in mind. As with the previous protocol, it is best to face the sun directly, not through a window or windshield, and without wearing sunglasses or a brimmed hat. Perhaps you can get an unobscured view by going up to a rooftop or by walking a few blocks from where you work or live. From a health perspective, that is well worth it. Again

## HOW IT WORKS

our circadian clock forward, making us want to go to sleep earlier and wake up earlier. Viewing late afternoon or early evening sunlight delays the phase of our clock, which, because it shifts the timing of the release of hormones like melatonin, makes us want to go to bed later and wake up later. So viewing both morning and afternoon sunlight thus helps keep us on a regular sleep-wake schedule, because the phase shift effects of morning and evening sunlight offset each other.

Another great reason to view sunlight in the late afternoon or early evening is that it helps buffer us against the negative effects of viewing bright artificial lights at night. In the morning, we need to see a lot of bright light to train our SCN with the solar cycle, and to elevate our levels of alertness. Unless it's a 10,000 lux lamp, most artificial lights and certainly our phone or computer screens just can't do the job at that time. After sundown, however, it takes extremely little bright light to quash our melatonin levels, and shift our circadian clock. And even if we can still fall asleep at night, that sleep is not going to be as high quality. In fact, studies show that as little as fifteen seconds of bright light can significantly shift our circadian clock. And while most screens are not as bright as the lights used in these studies, our ipRGCs and SCN work to sum photons—that is, add up all the light we see from various sources. And let's face it, most people are looking at their phone and other screens for a lot longer than fifteen seconds after sundown.

To recap: In the morning, you need more bright light in your eyes than a screen or conventional home lighting can provide, and at night you want far less light reaching your eyes than typically comes from a phone or computer screen or conventional home lighting. It all has to do with how our eyes are more sensitive to detecting light at different times of day or

night. If that seems inconvenient, you're right; nowadays it is! But the circuits underlying retinal and circadian sensitivity across the twenty-four-hour cycle evolved long before humans had artificial lighting, much less the bright, short-wavelength-enriched LED screens and LED bulbs that we use today. I, like most people, often wake up once in the middle of the night to use the bathroom. If I had things my way, our circadian system would be buffered from the phase-shifting effects of light at that time, but I wasn't consulted at the design phase, and our ipRGCs are exquisitely sensitive to bright light in the middle of the night and wee hours of morning. In fact, our ipRGCs and SCN become increasingly responsive to light with every hour after sundown.

If you're wondering if fireplaces, campfires, candlelight, or moonlight are too bright to view after dark, the answer is no. Firelight sources emit very long wavelengths, and moonlight is low lux. Candlelight is typically just 1 lux. A bright, full moon is less than 1 lux! Compare those numbers to a typical table lamp, which is 200 to 400 lux, or commonly used overhead indoor LED lighting, which is 500 to 1,000 lux. So are we to all avoid artificial lighting and opt for candlelight or moonlight? Maybe, but more on that later. In the meantime, let's acknowledge that most people just won't do that, and even if we opt to at home, we can't control the lighting of most other environments like office cubicles, restaurants, hotels, and the rest. That is where this protocol can help.

Viewing sunlight in the afternoon reduces the sensitivity of neural circuits in your retinas, so that any artificial light you see after dark does not have as strong an effect in disrupting melatonin and other circadian-related molecules and processes in your eyes and brain. This protocol is especially important if your mornings and days are mostly spent under artificial indoor lights, as that will cause any artificial lights you view at night to have an outsized negative effect on your sleep and circadian rhythms. Think of viewing the sunset as your (partial!) nighttime TV-watching or phone-scrolling inoculation. That said, you should still strive to dim the lights in the evening, and make it as dark as possible while you sleep, as discussed in the next protocol.

**FREQUENTLY ASKED QUESTIONS**

**What if I can't get myself to a place where I can see sunlight in the late afternoon or early evening?**

I understand. Most of us are working, driving, or otherwise occupied indoors when the sun is setting—especially in the fall and winter months. And while sunsets are both beautiful and beneficial, telling your boss "I

need to leave to go watch the sunset" is not going to go over well. (Although saying "Let's go watch the sunset together because it's good for our health" might be a good idea, depending on how you frame the question.) If you can't watch the sunset, just try to go outside at some point in the afternoon to get

▸ Avoid Viewing Bright Lights in the Final Hour Before Bedtime

## WHAT TO DO

• After sundown, limit your nighttime light exposure. You have a few options for how to do that. Which you use will depend on where you are and what activities you need to carry out.

• Option 1: Dim the lights. After sundown, dim the lights in your home or wherever you are in the hours before sleep.

• Option 2: Use red lights at night. I am a fan of using soft amber or red incandescent (or even amber or red LED lights, which are easier to find nowadays) for indoor lighting after dark as they don't spike cortisol or suppress melatonin as much as many bright white LED lights do. Remember, you want to try to keep your cortisol low in the late evening and especially the final few hours before your bedtime at night. You also want your melatonin elevated then. While this protocol might seem extreme, shifting to amber or red lighting in the final hours before sleep can

Note: I am not talking about "red light therapy," and you don't need a fancy red-light panel for this protocol. In fact, red-light panels tend to be very bright—exactly the opposite of what you want to use at night! They have their place, which I will discuss later in this book.

• Option 3: Use combination UV, blue, and green light blocking glasses after sundown. This is the most versatile and cost-effective option. Consider buying a pair of blue and green light blocking glasses if you need to be on a computer, look at a phone, or watch TV at night. They are also helpful when you can't control the lighting because you're out at a restaurant, in an airport, et cetera, at night. I helped design ROKA Wind Down glasses, which block the full range of blue and green light, are lightweight, and come in a variety of frames. They are excellent quality.

• Option 4: Use good ol' candlelight. Just make sure to blow it out before you go to sleep!

## HOW IT WORKS

Humans are not the strongest or fastest creatures on earth, but we are the best at building technologies. Every technology—the automobile, the smartphone, airplanes, antibiotics—brings the potential for unprecedented power over nature in our daily lives. Artificial lights have vanquished darkness. But in the case of LED lights, in particular, this technology also carries a risk to our circadian cycle, our mitochondria, and thus, our health.

Until about ten years ago, most artificial lighting used incandescent bulbs, which contain more long-wavelength light and a more even distribution of different wavelengths than LED bulbs. Incandescent bulbs have been largely phased out because they are less energy efficient than LED bulbs. Nowadays, most indoor lighting comes from LEDs, which emit a disproportionate amount of short-wavelength light that can potently reduce melatonin levels. Emerging evidence also points to negative effects of LEDs on mitochondrial health (which we'll go into later, in chapter 5). Those are some of the risks.

Now let me explain why bright artificial light at night may also be affecting our mood. Each of us has a brain structure called the habenula that sits at the middle and top part of an egg-shaped brain structure called the thalamus. Your thalamus is a station for routing different types of sensory and motor information to the correct parts of your brain so you can detect and integrate that information into your thoughts and actions. When I was an undergraduate and graduate student taking neuroanatomy, the habenula was listed as having an "unknown function." Recent studies in rodents, fish, and humans finally revealed what it does. We now know the habenula is part of a network that controls depressive feelings and disappointment. Habenula activation creates those states by altering dopamine and serotonin dynamics in brain areas associated with reward. Why would we harbor such a structure and circuit? The answer is that not all behaviors are meant to be rewarded. Some need to be punished so we are less likely to repeat them. And it turns out that viewing bright light in the "you really should be asleep now" phase of your circadian cycle is one of them. Animal studies and emerging evidence in humans show that viewing bright lights at night causes your ipRGCs to activate a specific region called the perihabenula. That leads to elevated levels of cortisol, reductions in dopamine, next-day anxiety, and deficits in learning—all consequences that appear to be separate from any reductions in total sleep time.

For most people, the "you really should be asleep now" time is somewhere between the hours of 11:00 p.m. and 4:00 a.m. But of course, neither your SCN nor your habenula knows clock time; they only know circadian time, which follows the circadian body temperature, cortisol, and other twenty-four-hour rhythms I described briefly earlier in this chapter. Your body temperature starts to rise in the final two hours before waking, and continues to rise throughout your morning and day until you reach the mid to late afternoon. At that point, it starts to decline. That decline continues as we sleep, reaching its lowest point, called, unimaginatively, your temperature minimum time about two to three hours prior to waking. So when I say avoid viewing bright lights from LEDs or other sources at night, I am referring to the time about six to eight hours before your temperature minimum time. This may seem overly technical, but knowing your temperature minimum time can be very valuable. And no, you don't have to measure your body temperature; your temperature minimum is the approximate time in your twenty-four-hour circadian cycle that roughly corresponds to two hours before you typically wake up, alarm clock or no. The

A bit more on those mood improvements: In 2018, a team of researchers, including noted UCLA psychiatrist Peter Whybrow, reviewed the data on the potential link between LED lighting and mental health. They concluded that exposure to blue light, in particular, posed a risk to sleep and other processes governed by the circadian system. There is also evidence suggesting that the way in which artificial lighting at night interferes with this system may lead to depression.

These findings do not mean we should all throw away our LED lights. However, if you can find incandescent lights for your home, they are preferable because they're less disruptive to the circadian system. In a study from 2020, researchers found that incandescent lighting interfered with the production of melatonin far less often than LED or fluorescent lights. (I cover why this is in chapter 5.) If you can't find incandescent bulbs, consider the Bon Charge or Ra Optics bulbs or ROKA Wind Down or other short wavelength (not just "blue") blocking glasses. Experiment with the options in this protocol to see what works best for you. And no matter what bulbs you choose for your home, still dim them before bed, and keep it dark in the six to eight hours prior to your temperature minimum time. If a light is bright enough, it doesn't matter if it is an LED or incandescent, it will still alter your circadian cycle.

**FREQUENTLY ASKED QUESTIONS**

**What if I want to go to a game, concert, party, or movie and view bright lights? Will that destroy my sleep that night?**

This is an important question, as we need to live a life that is not just focused on health. From a circadian perspective, no, it will not destroy your sleep. Your natural wake-up and to-sleep times are determined by your genetics, your social activity, your exercise routine, your stress levels, and the average timing and amount of your morning and evening light viewing—especially morning sunlight—in the prior days. As I mentioned earlier, the idea is to strive to get great sleep about 90 percent of the nights of your life, and reserve the remaining about 10 percent for when challenges or fun keep you up late, or even until dawn.

Yes, wearing glasses that block UV, blue, AND green wavelength light can help keep your sleep on target, even when you're going out a lot, but who wants to wear those lenses on a first date or on your wedding night? They can be fun to wear to concerts and dinner, but I hate to think that anyone would miss out on the joys of life to avoid bright light exposure to their eyes. That said, if you're getting a lot of screen LED light in your eyes at night because you're scrolling on your phone or computer or watching a big flat-screen monitor, that will take a negative toll on your health. Please adjust this protocol so that it works for you. Also, you shouldn't wear these glasses while driving, for obvious reasons; there are a lot of road signs and lights that you would not be able to see accurately—or at all depending on their color, and you could get into an accident.

What I describe here is what to strive for. Everyone diverges from that now and again. Enjoy your life and meet its moments by prioritizing your health most of the time and not stressing about the occasional night out.

**What about kids? Do the same rules about lighting apply?**

Yes. They need great sleep even more than we do, given the stage of brain and bodily development they are in. Teens, though, tend to stay up late to socialize online or otherwise. Keep in mind that kids and teens often need eight to ten hours of sleep. But their socialization is vital, too, so you'll have to strike a balance between encouraging a healthy relationship to artificial light, screens, and their social life. That said, if your teen or child is depressed or anxious, it's important to assess their sleep patterns and relationship to light, especially the amount and timing of screen use. I have heard from many parents whose kids had a significant improvement in their mental health with the protocols listed here. Some were only able to convince their teens to use their phones outside in the yard instead of in their rooms in the morning. Others have told me their kids love the

morning and evening sunlight viewing for the immediate mental and physical benefits it provides. I won't make this a parenting protocol, but the phrase "be a channel, not a dam" comes to mind as possibly useful in this context.

**My infant wakes up frequently throughout the night. Can dim lighting help me cope with interrupted sleep?**

Yes. Even if your infant is waking frequently through the night, try to keep the lights as dim as possible during those stretches, and/or use red light bulbs (again, not the same as red light

▸ Find Your Ideal To-Sleep and Wake-Up Times and Strive to Adhere to Them

## WHAT TO DO

• Find your chronotype to set your schedule. Some people sleep better when they stay up late and sleep in late; others feel best when they go to bed early and wake up early; and most people fall somewhere in between, preferring to go to sleep somewhere between 10:00 p.m. and midnight and to wake up between 6:00 and 8:00 a.m.

• Try to stick to your schedule. As long as you sleep for at least six to eight hours a night, pick a sleep-wake schedule that works for you. Even shifting slightly closer to your chronotype will afford you much more restorative sleep.

A systematic approach to figuring out your ideal sleep-wake schedule:

noted that the intensity of the light the participants experienced while camping was four times greater than the intensity of light in their normal lives—a testament to how much time they normally spent indoors. Interestingly, just one weekend spent camping had a similar effect of shifting people's circadian rhythm toward being early risers and people who prefer to go to sleep early. Mind you, those shifts were not just while camping. They persisted after they came back. This study was small and in no way does it mean there is no such thing as a night owl. But the rapidity and degree of the change suggest that artificial lighting masks the preference of many people to have a solar-cycle-aligned rhythm.

There are potential health implications tied to shifting toward (or away) from your innate sleep-wake timing. And they suggest that except for bona fide night owls, a more solar-cycle-aligned rhythm is beneficial. A large-scale 2019 study of nearly 700,000 subjects showed a causal link

between being a morning person and better mental health—that's right: a causal link. A different study found a higher risk for cardiovascular disease among people who went to sleep later and slept for a long time. Another analysis using data from 6,258 people in Finland found a link between routine late nights and type 2 diabetes. The patterns a night owl keeps have also been linked to depression. And there are new data showing that even if you prefer to go to bed and sleep late, shifting to becoming an early riser can improve your health. But telling a night owl to become a morning person can be dangerous—people cling to their natural sleep cycles like a pit bull on a rope toy.

Whether you think you're a night owl or not, by systematically experimenting with your to-sleep and wake-up times a bit and by limiting bright artificial light exposure in the final hour or two before sleep, you can determine the ideal sleep-wake schedule for you. Few people can do genetic testing or sleep lab analysis to determine their true chronotype, but there is a simpler way and it's worth exploring. Here's how: Unless you are perfectly happy with your sleep, for one week, try getting into bed by 8:30 to 9:00 p.m. If you're not ready for bed at that time, read a book or do another nonscreen activity, but turn all the lights out by 9:00 to 9:30... maybe 10:00 p.m. at the latest. Many people who do this wake up at 6:00 a.m. or even 5:00 a.m. the next day feeling more refreshed than ever before. Those who had previously been going to sleep at 11:00 p.m. or 12:00 a.m. and assumed they needed eight or more hours of sleep to feel rested often find they can feel great with just six or seven hours of sleep if they just get to sleep a bit earlier. This process aligns your sleep more accurately with your genetically driven cortisol and melatonin rhythms.

The genes that determine chronotype impact what are called clock genes, which are located in your SCN, and in almost every other cell of your body too. Your SCN entrains every cell in your body by changing its patterns of clock gene expression, either directly or indirectly, by altering the timing and type of peptides, hormones, and neural signals those cells receive. It's a multipronged and highly distributed network that allows your SCN to do that. That is one reason why once people find their correct sleep schedule, they often notice their digestion and immune systems improve, as well as key biomarkers of metabolism, not just their sleep and mood. It's not unusual to get more deep and REM sleep and awaken feeling substantially more refreshed when you obey your chronotype. Nocturnal growth hormone release is strongly regulated by your circadian clock, and is released in the first hours of sleep. If you are meant to go to sleep between 8:30 and 10:00 p.m. and you stay up late, you miss out on getting some of

that growth hormone. And growth hormone is vital for bodily and brain repair, metabolism, and myriad other health functions.

A few things to keep in mind as you try this protocol. If you typically have been going to sleep after 10:30 p.m., dim the lights as described in Sleep Protocol 3. Also, remember that it is perfectly normal to wake up once to go to the bathroom in the middle of the night. If you do, I highly recommend that you do not look at a clock or turn on bright lights. Chances are you'll fall back asleep within thirty minutes or less. No matter when you typically have been going to sleep, eating too close to bedtime and/or drinking caffeine late in the day will mess up this whole experiment—both lead to more frequent nighttime wakings, so don't do either. In general, eating your last bite of food at least two to three hours before bedtime is best, provided you are not experiencing gnawing hunger near bedtime, in which case a small snack one hour before is helpful. Yes, eating too close to bedtime may disrupt sleep, but gnawing hunger will keep you awake, so we need to be practical about this whole "don't eat too close to bedtime" thing. And try not to drink caffeine after 1:00 or 2:00 p.m. (more on the reasons why in Sleep Protocol 11).

If you are not used to going to sleep this early, and you find yourself unable to fall asleep even after several nights, you may just be genetically wired to go to sleep between 10:30 p.m. and midnight. True night owls do exist! There is nothing wrong with that, provided your schedule lets you sleep long enough for you to feel rested the next day. Again, this assumes you are not drinking caffeine after 1:00 or 2:00 p.m., stressed out, or otherwise generating elevated levels of arousal late in the day, all things that can affect your sleep-wake schedule.

Indeed, people prefer different sleep schedules. Approximately 33 percent of people feel great going to sleep between 10:30 p.m. and midnight and waking up between 6:30 and 7:30 a.m. The other 33 percent do best going to sleep between 8:30 and 10:00 p.m. and waking up between 4:30 and 6:30 a.m. The remaining third are bona fide night owls—in the absence of caffeine or work demands, their genetically driven melatonin rhythms and circadian rhythms cause them to be wide awake until 12:30 a.m., 1:00 a.m., or even 2:00 to 3:00 a.m., and they do best waking up between 9:00 and noon. The point is to determine which group you are in, and to do so while not letting other variables screw up your experiment.

It is oh-so-valuable to know what your body and brain need, and thrive on. If you're truly an early or late chronotype, perhaps you can delegate tasks to family members or roommates to meet life's demands. You can select

courses and workout times that better match your rhythms. I will even go so far as to say that, because of the way that getting your sleep-wake times right generates outsized positive effects on your health and cognition, you might even be wise to try to build your life around it. Tell your boss you'll have fewer sick days (you will), tell your partner you'll be a better partner (you will), and if you have kids, a better parent (you will). Feel free to cite me during the negotiations.

There are aspects of our biology that we can neglect for a while before it causes issues. Chronotype is not one of them. Of course, we can shift our circadian clock for child-rearing, work, or travel, and this book teaches you how to do that more effectively, but we can't disobey our chronotype for more than a few years before it catches up with us, and often it only takes a few weeks of disobeying it before we start to notice mild to severe malaise, irritability, or poor digestion. Fortunately, the reverse is also

—also may happen; it's well documented in sleep studies, although to my knowledge its physiological basis has not yet been explored in detail. Remember, this is a transient increase, so don't sweat it. Let it pass and embrace the drop in body temperature, the rise in melatonin, and sleepiness that follows.

## HOW IT WORKS

The intricacies of what happens when we sleep—the so-called architecture of sleep—deserves a book of its own, and indeed many have been written about it. Rarely discussed, however, is how the architecture of each sleep-wake cycle influences our sleeping and our waking states. We touched on this topic a bit earlier in this chapter. But now, by exploring the architecture of sleep itself, we can start to understand how the placement of six to eight hours of sleep in each twenty-four-hour cycle, and the consistency of that placement from one day to the next, has a profound effect on exactly what happens while we sleep, and how that impacts whether our sleep is maximally restorative.

### The Stages of Sleep

Many people are familiar with the two major types of sleep: rapid eye movement, or REM, sleep, and non-REM sleep. Non-REM sleep is often referred to as deep or slow-wave sleep, but that's oversimplified. Non-REM sleep consists of three distinct phases that we enter, one after the other. These are referred to as N1, N2, and N3 sleep, and only in the N3 stage does actual deep or slow-wave sleep occur. Cycling through N1, to N2, to

N3, to REM sleep takes, on average, about ninety minutes. So a person who sleeps for three hours might make it through two cycles, a person who sleeps for six hours experiences four cycles, and a person who sleeps 7.5 hours gets five cycles, and so on.

We enter N1, or light sleep, first and it typically lasts five to ten minutes. It is easy to rouse us from N1 sleep. But N1 is not merely a transition stage; during N1, our body starts to generate additional adenosine triphosphate, or ATP, the chemical that fuels all our biological processes. What's more, proteins that have become misshapen or broken during our waking hours get repaired during N1, and proteins that help build memories by reconfiguring our neural circuits are created.

Next we enter N2 sleep, which lasts about twenty minutes. During N2 our body temperature drops, our heart rate slows down, and our brain produces sleep spindles—surges of transient brain activity that help consolidate information we have learned in the days prior. Sleep spindles are vital for learning new cognitive information and motor skills, which both represent types of neuroplasticity. We explore this in greater detail in chapter 6. N2 has another intriguing feature: PGO waves, which are surges of large-scale brain activity that travel from the pons (P) of the brainstem, to the geniculate (G), a structure in our thalamus (the sensory waystation we discussed earlier), and then to our occipital (O) cortex, which resides in the back of our brain and is involved in, among other things, visual processing. PGO waves trigger the release of massive amounts of glutamate, an excitatory neurotransmitter that is a requisite component of the multistage process of learning, such as in the formation and elimination of synapses.

After N1 and N2 we enter deep sleep, what sleep researchers call slow-wave sleep, or N3. This is an extremely restorative stage of sleep. During N3, our brain waves shift to very high amplitude and low frequency, aka slow; our heart rate slows way down, and growth hormone is released from our pituitary gland into our bloodstream. What's more, the active repair of our muscle, brain, and other bodily tissues occurs—including our tendons, skin, heart, et cetera. That process involves a lot of protein synthesis, or the repair and regeneration of cells and tissues by use of specific amino acids. Some of that repair and regeneration occurs in response to tissue damage from exercise, but most of it is simple "housekeeping"-based cellular turnover and repair, which is also vital to our health and longevity. N3 is also when glymphatic washout of our brain occurs. This process was discovered just in the past fifteen years or so and involves the pulsing of lymph-like fluid through our brain, thereby

removing the waste products caused by normal brain metabolism and any cellular debris that accumulated the previous day. This glymphatic clearance is actively governed by multiple brain regions—including the locus coeruleus—that we will talk about more below and in later chapters.

REM sleep, or rapid eye movement sleep, follows N3, but often we cycle from N3 back to N2 for a few minutes or more before advancing into REM sleep. REM sleep is crucial for sorting through all that waking life brings our way by eliminating or strengthening specific neural connections in accordance with what information we need to keep and what we should discard. Thus, think of REM sleep as an experience audit, with only the meaningful experiences surviving the cut.

One more interesting thing about REM: Our body becomes mostly paralyzed during this stage, a phenomenon known as sleep atonia. The most prevalent hypothesis to explain sleep atonia is that it keeps us from acting out our dreams during REM sleep. Our brains are extremely active during REM and the dreams we experience during this stage tend to be vivid and emotionally laden—often marked by flying, swimming, being filled with anxiety or awe, and highly dynamic in how we move through time and space (in the dream of course). The idea is that REM-associated atonia prevents us from acting in accordance with what are essentially in-sleep hallucinations, which would put us in danger.

The relative amount of time we spend in REM sleep increases, and the amount spent in deep N3 sleep decreases, as we complete more sleep cycles across a night of sleep. This means our most vivid dreams tend to occur as we approach morning. A baby's sleep is largely REM sleep, and the amount of REM we experience decreases with age starting in adolescence. If we are sleep-deprived, or we do something that reduces the amount of REM sleep we get on a given night—using cannabis, for instance—the next night we experience REM rebound by entering REM sleep sooner in the sleep cycle, and experiencing more REM overall. Chronic cannabis users who stop using cannabis

## Health risks of sleep deprivation

Given all this, it becomes clearer and clearer how lack of sleep, but also altered sleep architecture, affects our mind and body in a variety of ways. Sleep deprivation is associated with diminished response time, alertness, attention, memory, and executive functioning. Additionally, as described at the chapter outset, lack of sleep can bring about longer-term problems as well. Insufficient REM sleep may increase the risk of dementia. In 2017, a

group of researchers published their investigation of links between dementia and sleep architecture. They looked at three years of data on thirty-two adults over age sixty with dementia and found that the likelihood of cognitive decline fell in line with a reduction in REM sleep. In fact, with every 1 percent reduction in REM sleep, the risk of dementia rose by about 9 percent. (Non-REM sleep deprivation did not share this connection.) The role of REM in building and consolidating neural connections, thereby protecting against cognitive decline, may also explain why truncating this stage would raise the risk of dementia, but direct evidence for this mechanism in humans is still lacking.

Getting sufficient REM sleep is also connected to overall longevity. In 2020, a group of researchers from multiple institutions reported that less REM sleep is associated with an increased mortality risk. Using data from 4,050 people, they found that for every 5 percent decrease in REM sleep, the mortality rate increased by 13 percent. That held true for most causes of death, including cardiovascular disease. Ask any sleep researcher "How can I get more REM sleep?" and they will tell you "Sleep in an extra hour or two." But most people can't afford to sleep in an extra hour or two past their usual wake-up time.

The good news is that when we sleep is almost as important as getting enough of it. If we get to bed and wake up at a time that is in sync with our chronotype, we end up getting more growth hormone release, and we spend less time in light sleep and more time in deep sleep and REM sleep. As a consequence, we tend to need less sleep overall. Our sleep can become more efficient and restorative.

Studies of shift workers tell us what happens if we sleep enough, but at the wrong times of the twenty-four-hour cycle. The short answer is: Bad things happen. Working at nighttime and sleeping in the daytime is associated with greatly elevated risk of type 2 diabetes, cardiovascular disease, depression, and suicide. Shift workers are also susceptible to neurocognitive impairment and at greater risk for substance abuse. One large survey from Canada found that executive function is significantly worse among people whose jobs take place outside normal work hours. One study found the level of performance impairment to be equal to a blood alcohol concentration of 0.10, which exceeds the legal limit for driving in many places. To be clear: Being a night owl is not the same thing as forcing yourself to be nocturnal, so if that's your chronotype, don't worry. The health detriments of shift work are likely to be multifactorial—the lack of sunlight, the altered social interactions associated with an inverted schedule, and more. We know that one of the

worst types of shift work for one's health is the swing shift, where people switch back and forth from daytime to nighttime work each week or even within the same week.

How long we sleep, when we sleep, and how consistent we are with our sleep-wake schedule affect our physical and mental health. That's why I hope you try the approach outlined here. Figure out what is best for you and then stick with it.

## FREQUENTLY ASKED QUESTIONS

### Does changing time zones for travel or moving the clock for daylight saving time affect our sleep-wake cycle?

Yes. When we turn the clocks forward an hour to move from standard time to daylight saving time, or back an hour to return to standard time, or when we change time zones during travel, our circadian clock has to shift. Changing our clocks forward or back means we have collectively decided that 7:00 a.m. is now 8:00 a.m., or 8:00 a.m. is now 7:00 a.m., but our SCN really doesn't care about clock time, it only knows time according to the entrainment cues it receives. You already know the most powerful entrainment cue and that's viewing morning sunlight. Even if we move the clocks just by one hour, the SCN needs a few days to shift its patterns of gene expression and communication with all the cells of the body to get them on schedule. In the meantime, we are groggy at times in the morning when we were perfectly chipper the day before. And a whole host of other systems in our body and brain have to shift

celebration and don't despair about arriving home late. One late night is not something to worry about.

Many people wonder whether wake times should be later after that occasional late night. If you usually go to sleep somewhere between 8:30 and 11:00 p.m. and have a night where you're up until 1:00 or 2:00 a.m., should you sleep late or wake up at your usual time? In general, it's best to get enough sleep to feel rested but not to sleep more than one or two hours past your usual wake time. So on those "I stayed up too late last night" mornings, get an extra hour in bed asleep if you can, and try to take a twenty-minute nap, or practice non-sleep deep rest (NSDR) in the afternoon (more on that soon), and then get to bed around your usual time that night. Your SCN—and thus you—thrive on consistency. If you go to bed much later than usual and sleep in much later, too, be sure not to shift your whole schedule forward. It's still fine (and encouraged) to view sunlight, exercise, and so on, but be especially mindful not to drink caffeine

much later in the day than usual, or you will soon find yourself off your usual rhythm and it will take three to four or more days to get back to it.

**What about children and teenagers?**

The advice is a little different for young children and teenagers. Babies need to sleep A LOT and enter REM far more frequently and for longer than any other age group. Toddlers and young children should also sleep as much or as often as necessary, or desired, to support their extraordinarily fast brain and bodily growth rates. The amount of neuroplasticity in infants and children, all the way up to adolescence and beyond, is incredible—and sleep is a key part of that process. Limiting their sleep is never recommended but of course once kids are in school, setting a regular schedule (and keeping devices out of their rooms near bedtime and throughout the night) becomes important.

Sleep needs are somewhat different for teenagers. Bedtimes shift during puberty for at least two known reasons. First is what researchers call the homeostatic sleep drive. This term refers to a brain and bodily pressure to sleep that builds up gradually throughout the day, mainly due to adenosine accumulation. The longer we are awake, the more adenosine in our bodies, and the greater the pressure to sleep. Adenosine levels drop while we sleep and begin to mount again when we wake up for the new day. This pressure to sleep is separate from the circadian clock mechanisms we have discussed a lot thus far. Teenagers (good rebels that they are) resist this pressure and force themselves to stay up late, building up more adenosine and, therefore, the desire to sleep in the next morning. And they often don't need to force themselves to stay up late; it just happens naturally, as the hormonal testosterone, estrogen, and progesterone surges of puberty shift their circadian clock to a later phase. Melatonin production begins later in the teenage brain than in the brains of either adults or younger children. All of these changes leave the typical sixteen-year-old awake until about 11:00 p.m. And this age group should be sleeping about nine hours per night, making 8:00 a.m. the preferred wake-up time.

Studies have found that moving school start times to 8:30 a.m. or later leads to better grades and other measures of academic performance. There is even evidence suggesting that this shift also reduces car crashes among teen drivers. It's beneficial for teenagers to have days when they can sleep as long as they want, even if that means staying in bed until noon. That doesn't mean they should never be woken up earlier on a Saturday or Sunday, but parents and caregivers do not need to worry about prolonged sleeping on the weekends. Teenagers, yes, reference this book if you need

leverage to sleep in and parents, yes, you still want to encourage them to get sunlight in their eyes once they get up, even though it is midday, and not low-solar-angle sun.

Current data indicate that teenagers in the US are in the midst of an epidemic of sleep deprivation that is likely having serious consequences for their health. Ideally, young people between the ages of six and twelve years should sleep at least nine hours per night, and thirteen-to-eighteen-year-olds should sleep at least eight hours per night. More than 70 percent of young people do not sleep enough, according to one large survey from the CDC. Anxiety, mood disorders, obesity, high-risk behavior, poor academic performance, and substance use have all been tied to sleep deprivation among adolescents. Good practices for teenagers include shutting off screens at least an hour before bedtime, keeping the phone (and the laptop, the tablet, etc.) out of the bedroom at night and the room cool and dark, having a consistent bedtime, and, ideally, sleeping for those eight to nine hours per night.

deliberately suppress whatever thought was most likely to intrude into their mind before sleep found this didn't help—predictably, the thought they were supposed to be suppressing just became more preoccupying. Instead, try these somewhat weird tricks:

Take a mental walk: Imagine walking from your bed to your mailbox and then down the street and to a nearby location, then home again by the same or different familiar route. Occupying a busy brain with an easy and engaging task can help stop ruminating thoughts. Mental walks like these are occupying but not stimulating. And they have the tendency to lull people off to sleep.

Move your eyes following the pattern described below: Warning: This is going to sound a bit out there, but it works well for many people, and studies are indirectly starting to suggest why. Lie down in bed and close your eyes. Keep your eyes closed and move your eyes slowly to the right then to the left, then up, then down, then counterclockwise and then clockwise. Then cross them somewhat by directing them gently toward your nose. Finally, do a big inhale and long exhale. Repeat one to three times as needed.

Told you this would sound a bit out there! This trick often helps people fall asleep, because to do so we need to stop thinking about our limb position, or proprioception. Our visual and vestibular systems, or balance, and proprioception systems interact, and the eye movements just described "confuse" those systems. A similar mechanism partially explains why gentle

slow rocking and pacing can help babies, and slow rocking of a bed (yes, such beds exist) can help adults improve their sleep metrics. We need more data to understand the mechanism behind this pattern, which is why it's not its own protocol, but it is reasonable to assume it is related to the eye-to-vestibular circuits involved in, among other things, proprioception and sleep. New eye mask and rocking-bed technologies that trigger this eye movement pattern are designed to cause similar eye movements. I have tried a few of these nascent technologies and confess I fell asleep in less than two minutes! Meanwhile, this technique is free and easy to try. "Next thing I knew it was morning!" is not an uncommon refrain from people with whom I have shared this trick.

▸ Lower Your Core Body Temperature at Night

## WHAT TO DO

• Lower the temperature of the room you sleep in at night. Cooler ambient temperatures facilitate sleep. Keep your bedroom between 65 and 68 degrees Fahrenheit, and a maximum of 70 degrees.

• Warm your extremities before bed. A hot shower, hot bath, sauna, or even washing your face with very warm water can help induce drowsiness by warming hands and feet and by triggering a drop in core body temperature in compensation for this heating.

• Make sure your feet and hands can be easily exposed from underneath your covers. The skin of these areas is a primary way we shed heat from our body. Exposing them to cool air is one of the most effective ways to decrease core body temperature. So untuck your sheets at the foot and sides of the bed.

• Avoid wearing socks while you sleep, especially if you tend to wake in the middle of the night. Socks can stop the natural cooling process that

## HOW IT WORKS

need the change in temperature, but other factors are key too. Research has found that the ideal ambient temperature while we sleep is between approximately 66 and 70 degrees Fahrenheit, which typically brings the microclimate immediately around our skin down to somewhere between 88 and 95 degrees Fahrenheit. To fall asleep our core body temperature needs to decrease by about 1 to 2 degrees.

A lower core body temperature facilitates various sleep-related biological processes you are now familiar with. First, research indicates that this

reduction may help initiate non-REM sleep, and our lowest body temperature is associated with deep sleep. Melatonin, a hormone that facilitates falling asleep, is released in sync with the nightly drop in body temperature. In one small study of twelve women who were entrained to the solar cycle, the secretion of natural melatonin at night occurred with a simultaneous decline in core body temperature. Other studies have observed a similar coupling.

The stillness of our body during early sleep also plays a role in further dropping our core body temperature. In sleep, our autonomic nervous system shifts from more sympathetic activity (or tone, as it's often referred to), which is related to being alert and able to generate movement, to a far more parasympathetic tone, which is associated with calmness, lower heart rate, and less or no bodily movement. All that combined causes our blood to flow more to our periphery, to our hands and feet, where it "dumps heat," which in turn leads to a lower core body temperature.

This means that warming up our hands and feet a bit before getting into bed could facilitate falling asleep. That may seem paradoxical: Why would warming up your periphery help at a time when cooling down your core body temperature is the goal? Because heating your periphery acts as a trigger. A hot shower or bath, or a sauna, warms the surface of your body. That causes the preoptic area of your hypothalamus—a brain area that works sort of like a thermostat—to trigger a compensatory response, lowering the core temperature in your body. This process makes it easier to fall asleep, a phenomenon known colloquially as the warm bath effect.

Once you get in bed, having your sleeping environment cool is also helpful. Many people sleep best in a cool room but with their whole body under loads of warm blankets. Other people run hot and like to sleep under just a sheet or a light blanket. Try out a few of these recommendations and pick what suits you best.

### In My Experience

I prefer a comforter and a weighted blanket, but stick my bare feet out from underneath them, which we know helps cool core body temperature. And I keep the room about 68 degrees Fahrenheit. Closer to morning, I find that I often bring my feet under the blankets (in my sleep!), which we know can increase the amount of time spent in REM in the final few hours before waking.

Temperature-controlled mattress covers like Eight Sleep make controlling your bed temperature across the night easy, and I use one. (They are a

podcast sponsor, and I am a scientific advisor to Eight Sleep). But with or without Eight Sleep, it is a good idea to use programmable room temperature controls to help create that cooler-at-start and warmer-toward-morning sleep environment. Experiment and find what works for you.

**FREQUENTLY ASKED QUESTIONS**

**Do daytime temperature changes affect nighttime sleep?**

Some data from animal studies suggest that a slightly elevated body temperature for a few hours in the afternoon may lead to more time spent in N3 sleep at night, when waste products from the brain and spinal cord are cleared away (among other processes). That's one reason why many people find that doing some moderate to intense exercise in the early or midafternoon improves their sleep, provided they don't consume a lot of caffeine before their workout.

**So should I avoid exercising at night?**

It's best to avoid intense exercise late at night, within three to four hours of sleep. High-intensity exercise raises our core body temperature, which is counterproductive to falling asleep. That said, some people can only exercise at this time, and many learn how to down-shift afterward with a hot bath or shower, and by using some of the tools covered in chapter 3, like deliberate long-exhale breathing. If you consistently find it difficult to fall and stay asleep after exercising hard at night, consider moving your more intense workouts to mornings and doing something lighter in the evening. A review of several randomized clinical

▸ Increase Your Core Body Temperature in the Final One to Two Hours of Sleep Before You Wake Up

## WHAT TO DO

• Program your heating system for the room you sleep in at night. If you have one that can be set to different times, program it to rise one to three degrees above your room temperature, starting one to two hours before your wake-up time.

• Keep clothing by your bed. If you wake up one to two hours before your preferred wake-up time or in the middle of the night to use the bathroom, try putting on some socks, sweats, or even a knit cap and go back to sleep until your preferred wake-up time. This can increase your core body temperature, which increases the amount of REM sleep you

get and will help you wake up feeling energized.

## HOW IT WORKS

We touched on this in the previous protocol

just have to get up and get to the thermostat. As mentioned before, devices such as Eight Sleep mattress covers that can cool at first and then heat your sleep environment in the final hours of sleep are very effective. These can also measure your deep and REM sleep, and use AI to make real-time adjustments to your sleep temperature to help you get the best possible sleep. But these carry a cost that for many could be prohibitive, and most versions don't travel easily. Anyone can adjust their bedroom temperature and/or use clothes to try to achieve some of the ideal temperature ranges described here.

▸ Nap (Correctly) to Offset Minor Sleep Deficits

## WHAT TO DO

• If you need a nap, make it short. Keep the duration to no more than about twenty minutes, and avoid naps longer than ninety minutes.

• Avoid late-day naps that interfere with your nighttime sleep. Each person can determine what time constitutes "too late" with a bit of experimentation. See how groggy you are in the hours following your nap. Most importantly, note if you can still easily fall asleep at your regular bedtime afterward.

• Know your goal. Nap strategies depend on what you are trying to accomplish.

• If you need an energy boost to sustain your focus and drive for the rest of the day: Aim for a fifteen-to-twenty-minute nap. Five to ten minutes will give you a mild energy boost and help your focus but it doesn't last long. After a fifteen to twenty minute nap, concentration, mood, and energy will improve in a way that will sustain you for the remainder of the day.

## HOW IT WORKS

getting less than optimal sleep at night, and you should focus on improving that instead. Others take the stance that napping is fine, beneficial even, and not always reflective of a sleep deficit. Provided you adhere to a few simple constraints, napping is great. But if you don't crave naps, there is no

need to nap. And yes, there is a proper way to nap.

First off, it's normal to feel a bit sleepy in the afternoon as your body temperature shifts from rising to falling—remember that the rate of core temperature dropping is what initiates the feeling of drowsiness. For people on a conventional schedule of being asleep at night and being awake in the daytime, this dip in early afternoon energy passes. I fall asleep at about 10:30 p.m. most nights and wake up at 6:30 a.m., and I always get a bit sleepy between 2:00 and 3:00 p.m. irrespective of what I eat, how much caffeine I drink, and so on. After 3:30 p.m. or so, I feel wide awake and energetic again. The point is, you might get a bit tired in the afternoon, but the feeling is temporary. Not everyone experiences this dip. Some people just barge right through the day, morning until night.

We know that napping can increase our ability to focus and learn. Matthew Walker, a professor of neuroscience and biomedical engineering at the University of Texas in Dallas, conducted a study in which a group of participants learned a new set of facts at midday. Half of them napped for ninety minutes afterward; half of them relaxed without sleeping. At 5:00 p.m., they learned a second new set of facts. The group that did not nap experienced a decline in learning capacity. The nap group sustained or even improved their capacity to learn. The difference reached about 25 percent, which is meaningful. The finding goes back to the role of sleep spindles, which we covered in Sleep Protocol 4. These are the types of brain waves present during N2 sleep, which happens early in the sleep cycle. Even a brief snooze of twenty minutes takes us into N2. Sleep spindles help us process new learning, but also prepare our brain for more learning after we wake up.

Napping also influences our mood. In another study, Walker and his group asked participants to rate the level of emotion they observed in a photograph. How angry were the angry faces? How happy were the happy faces? Again, half the subjects napped for about an hour and half did not. Toward the end of the day, the non-nappers were rating the emotions as more extreme; angry people were angrier, and so on. The napping group was more likely to rate faces as happy; the rest time seemed to blunt the normal increase in anger sensitivity across the day. As Walker sometimes puts it, sleep gives us a "rose tint" to our worldview. In this study, the naps were ninety minutes, long enough to get some REM sleep, so the benefit is likely due to that, which as we discussed, is when we unload the emotional burdens from our daytime experiences. Although, in this case, it wasn't about the nappers' emotions; it was about how they perceived the emotions of others.

Finally, there is the simple benefit of restoring energy. Many people experience what sleep researchers call a postprandial drop, a dip in energy after eating, especially large meals rich in starchy carbohydrates. Aside from modifying your dietary pattern, naps can help get you to the other side of a postprandial drop if and when you do experience the phenomenon. That said, for anyone who feels tired in the middle of the day and would do better without that sleepiness, I suggest making sure you don't overeat, and possibly even reduce starchy carbohydrate intake, at lunch. Take a short ten-minute walk afterward; that can help for reasons related to blood sugar, or glucose, regulation (covered in chapter 2). Many people find that just reducing the amount of starchy carbs they eat at lunch allows them to go straight through the rest of their day feeling energized and focused. I'll talk more about the relationship between food and brain states in chapter 4.

### When Not to Nap (and Why)

A short midday sleep isn't a good idea for everyone. To understand why, we need to return to adenosine. As you recall, adenosine acts as a neurotransmitter that inhibits neural activity in brain regions involved in strategic planning. Sleep actually results from two processes: the circadian clock and sleep pressure from adenosine. Most of the protocols thus far depend on the former. This one has to do with the latter, which you might remember from Sleep Protocol 4. Adenosine is so vital to this process that we ought to explore it a bit further, so here goes.

ATP, or adenosine triphosphate, is the basic unit of energy for all our cellular processes. It is generated during the early stages of sleep. Adenosine is the downstream product of the burning of ATP. It's a product of metabolism that accrues as we engage in activity over the

minutes and more than ten, so set your alarm for a fifteen-to-twenty-minute nap and perhaps hit the five-minute snooze button if you don't feel refreshed the first time your alarm goes off. Just don't hit it a second time! A twenty-minute nap shouldn't make you groggy. That feeling, known as sleep inertia, happens when we wake up in the middle of the deeper stages of non-REM sleep. By keeping the nap to about twenty minutes, you get the benefits of N1 and N2 sleep without suffering from sleep inertia.

Perhaps surprisingly, some people find that resting without sleeping is sufficient to restore mental and physical energy. When you do, you can enter a liminal state in which you are still awake but some parts of your brain are in a sleeplike state. Many people find that even if they just lie

down, elevate their feet on a few pillows or a chair, close their eyes, and rest for ten to fifteen minutes, they get some of the restorative effects of a real nap. Elevating your legs and feet slightly in this way promotes a parasympathetic response, increases venous blood flow to the heart, may lower blood pressure slightly, and—provided you don't ruminate on stressful topics—can shift your brain's state toward one of more "relaxation." No, it's not a good idea to spend this time tapping or scrolling on your phone, as it will offset the brain state shift you're seeking. You need to get your mind into a sort of idling mode.

Idling is clearly not neuroscientific language. But it's a good way to describe how our brains shift to a very different pattern of activity under these resting conditions, moving from higher frequency beta waves (13–30 Hz) to lower frequency alpha waves (8–13 Hz). Napping, on the other hand, lets our brain enter N1 and N2 sleep, allowing even lower frequency theta waves (4–8 Hz) to emerge. Experienced meditators can get themselves into a state where theta waves emerge quickly, although that takes practice. Relaxation techniques like those offered in the very next protocol can help you rest during the day without interfering with your ability to fall asleep at night.

**FREQUENTLY ASKED QUESTIONS**

**I'm not a napper but I don't have trouble falling asleep at night. Should I nap?**

If you are not tired in the daytime, don't force yourself to nap. Some people get through a day just fine without naps.

**Can naps help with insomnia?**

People with insomnia should generally avoid naps. The temptation to sleep during the day can be powerful when coping with poor sleep at night. But all evidence points to the idea that resisting the urge to nap is the way to go for insomniacs. If you are suffering from chronic insomnia, try the protocols in this book, and if those don't solve the issue then talk to a doctor who specializes in sleep medicine.

**For older adults, is there an added health benefit to napping?**

Studies of napping among older adults have shown mixed results. A concern with prolonged midday rests among older adults is that the need to nap may be indicative of a problem with nighttime sleep. Older adults who always feel tired during the day should consider changes that can benefit sleep, such as making sure to get plenty of natural daylight in their eyes in the morning and dimming indoor lighting at night before relying on

naps (Sleep Protocols 1–6 all have useful tools for this situation).

▸ Use Non-Sleep Deep Rest (NSDR) to Offset Sleep Deficits (and to Get Better at Falling and Staying Asleep at Night)

## WHAT TO DO

• Do non-sleep deep rest (NSDR) if you feel tired and need to restore your mental and physical vigor. NSDR is fast, safe, effective, and costs nothing. It has both immediate and long-term benefits for your brain state, and there is evidence it can improve focus and learning. It takes as little as ten minutes, although longer versions of twenty, thirty, or even sixty minutes can be even more effective. It's nearly impossible to do NSDR incorrectly (unless you are texting in the middle of a session).

• Do NSDR first thing in the morning if you failed to get enough sleep the night before and want to feel more rested as you enter your day. If you don't have time just then, I suggest you do NSDR in the midmorning or afternoon. It will still help.

## HOW IT WORKS

memory and improve mental health. When done consistently, meditation techniques can lead to profound shifts in our awareness of our internal thought patterns. And over time, meditation can provide a rich understanding of how our minds work and can even change our temperament.

Self-directed relaxation protocols like NSDR are different. NSDR involves shifting our autonomic nervous system into a more parasympathetic, or relaxed, state by breathing in a manner in which the duration of our exhales is deliberately emphasized. NSDR also involves deliberately and systematically moving our focus of attention from one part of our body to another, then to locations far from our body, then closer to us, then away again. The focusing, then defocusing, and moving and resetting of attention over and over keeps the mind active. And because it is done while the body is deeply relaxed, it brings about a somewhat dreamlike state, and yet you are completely aware and alert.

NSDR is a modification of an ancient practice known as yoga nidra, which roughly translates to "yoga sleep." Yoga nidra has been described as creating the experience of moving our mind away "from thinking and doing toward a state of being and feeling"—which is an admittedly vague statement. There is a small but growing body of modern science on the

brain and bodily states that yoga nidra and NSDR bring about and their ability to restore mental and physical vigor, improve consolidation of newly learned information, offset sleep loss, and more.

When I first tried yoga nidra in 2017, I found it to be incredibly restorative. Doing this technique for twenty to thirty minutes first thing in the morning after a restless or less-than-full night's sleep left me feeling fully rested, like I'd had the best sleep of my life. I also noticed that incorporating this practice in the morning improved my ability to fall back asleep quickly if I happened to wake up in the middle of the night. That's right—even though NSDR stands for non-sleep deep rest, and even though it is typically done in the morning or afternoon, something about doing it at that time significantly improved my ability to self-direct my brain and body into a relaxed state and drift back to sleep at night if ever I needed to. What's more, if I ever woke up in the middle of the night, listening to a guided NSDR recording beat tossing and turning, and often sent me back to sleep.

Challenges with falling asleep and staying asleep are very common, so I was intrigued. I also noticed that replacing an afternoon nap with an NSDR session left me feeling especially reinvigorated and did not alter my nighttime sleep in any negative way. I wondered how this simple practice really worked. So I dove into the literature and found there wasn't much known. But the few studies I did find were interesting. A 2002 study from Denmark found that doing yoga nidra dramatically increases levels of dopamine in a brain region called the basal ganglia, which are involved in taking, and withholding, action. Yoga nidra also reduced people's blood pressure and heart rate. Although this and other studies of yoga nidra were small in terms of the number of subjects studied, other investigations indicated that this practice could improve various measures of mental health and help consolidate certain forms of learning.

At one point, I decided to devote a portion of my lab to studying yoga nidra as a tool for stress reduction. To do so we needed to keep its core elements—breathing that emphasizes long exhales to slow the heart rate and place us into a more parasympathetic state, the deliberate and systematic movement of one's perceptual attention to the surface and then the interior of the body, then away from the body, and so on—but we opted to remove the "intentions" and any language pertaining to more spiritual aspects of yoga nidra. This parsing was not an attempt at cultural appropriation of an ancient tradition. Rather, it was a way to examine this powerful tool within the context of Western science, where we seek to understand what works and why, and to then use this knowledge to build more efficient and effective tools. Any aspect of the practice that couldn't

be reliably measured was discarded.

Later, as I moved into public education and spoke about stress mitigation tools, I coined the term non-sleep deep rest, which keeps the key elements related to the physiology of yoga nidra, but removes the subjective aspects that are hard to keep constant from one person to the next. And eventually, given the wide interest in stress mitigation tools, my podcast team produced various NSDR scripts of short (ten minute), medium (twenty minute), and long (thirty minute) durations and made them freely available online. Public feedback has confirmed the positive effects of NSDR, but of course it just follows the thousands of years of people across the world

**FREQUENTLY ASKED QUESTIONS**

NSDR to reset during the day, or first thing in the morning to offset the effects of not getting enough sleep, just be sure you set an alarm in case you fall asleep.

**How long should my NSDR session be?**

That's up to you. A ten-minute NSDR is often enough for a quick afternoon reset, but I suggest doing a twenty- to thirty-minute NSDR if you have time. If you slept poorly the night before, you may find that a longer session in the morning or afternoon leaves you feeling more rested, compensating for the sleep you missed in the night. Find what works for you.

**Can NSDR truly replace sleep or reduce the amount of time I need to sleep?**

Claims that "thirty minutes of yoga nidra is worth two hours of sleep" abound. I've also heard that "twenty minutes of meditation twice daily reduces your nighttime sleep needs by several hours." I am not aware of much laboratory data supporting those statements. It can be hard to get enough sleep, and most of us would love to need less of it to feel fully rested. NSDR won't necessarily reduce the amount of nighttime sleep you need, but it may very well improve the quality of the sleep you get. Combined with proper placement of your nighttime sleep in the twenty-four-hour cycle (see Sleep Protocol 4 for more), and other protocols in this chapter, it may, however, reduce your total sleep needs by improving your sleep architecture, as we discussed earlier in this chapter.

In one study, researchers randomized forty-one people with chronic insomnia to participate in either cognitive behavioral therapy for insomnia or twice-daily yoga nidra for twenty minutes per session. In both groups, participants slept a few minutes longer during the study than at the start of it. However, the yoga nidra group fell asleep faster, fell back asleep faster

after night wakings, and had a more efficient sleep (an increase in the number of times they went through N1, N2, N3, and REM) compared to their sleep prior to the study. So while they weren't sleeping less, the sleep they were getting was better than before. The study also found a significant reduction in cortisol following yoga nidra. That may be one factor that reduces the total amount of sleep you need because, in theory, lowering your evening cortisol should improve sleep. If you practice relaxation techniques during the day and happen to need less sleep that night, that's an added bonus.

▸ Consider Supplements to Support Sleep

## WHAT TO DO

• Find a combination of supplements that support you in getting the best possible sleep. Learn a bit about how each of these works. Consider your specific needs and consult with your physician to ensure that any supplements you take are safe for you and will not interfere with any medications you are currently taking. As stated in the introduction, supplements can have their place but ideally, try behavioral tools first. There is simply no pill to replace poor sleep habits, or "sleep hygiene."

Below is a list of common supplements sold over the counter that can assist in getting the best possible night's sleep:

• Magnesium L-threonate or magnesium bisglycinate. Dosage: 145–250 mg. These two forms of magnesium are interchangeable for improving sleep.

• Apigenin, a derivative of chamomile. Dosage: 50 mg.

• Theanine. Dosage: 100–200 mg.

• 100–200 mg theanine. (Don't take theanine if you tend to have intense dreams or nightmares, or if you sleepwalk, as it can enhance the vividness of your dreams.)

• Pay attention to potential side effects. The supplements listed above are generally very safe but not all of them will sit well for all people. Here's what to watch for:

• Gut discomfort: A small percentage of people (about 5 percent in my experience) who take magnesium threonate as a sleep aid experience digestive issues or other gut discomfort. Magnesium L-threonate and bisglycinate are far less likely to do this than other forms of magnesium like citrate or malate, et cetera. Still, discontinue use of magnesium

L-threonate or bisglycinate if gut discomfort persists. The dosage listed above for magnesium is already pretty low so you don't have a much lower limit to work with.

• Vivid or disturbing dreams: A small percentage (about 5 percent) of people who take theanine report having vivid dreams that wake them at night.

• Proceed with caution. Again, it is important to consult with a physician before you take supplements. You need to be sure that you are not risking a dangerous interaction, and that a medical professional has information about your sleep issues that may be symptoms of an underlying health problem. And always check that any product you decide to use is reliable.

## HOW IT WORKS

When it comes to getting optimal sleep, it's always best to start with behavioral tools and changes to your sleep environment. (Sun)light exposure, adjusting your core body temperature, and NSDR can all be employed to improve sleep. So, too, can exercise and proper nutrition (more on both later). These carry no cost, cause no side effects, and quite often resolve persistent issues.

Many people turn to supplements to help improve their sleep. Parsing the scientific literature on the efficacy of over-the-counter, supplement-based sleep aids can be difficult. The studies are often small, they have conflicting results, and they are too few to provide a significant body of evidence. That being said, enough data exist to indicate that some supplements may be beneficial and carry a very low risk of harm for most people trying them. In my opinion, these are much safer than the majority of prescription sleep medications.

A major issue when it comes to supplements is that the dose of the pills inside the bottle often does not match the dosage stated on the label. In addition, just because a supplement contains a "natural" ingredient does not mean it is safe. And these ingredients may react badly with a prescribed medication. So a reminder here, again, to talk to your doctor before proceeding, and to only buy supplements from reputable brands. I discuss a few things to look for on the packaging to ensure this below.

• Magnesium L-threonate: Increasing brain magnesium with this compound may improve learning and memory. A randomized study of magnesium L-threonate versus placebo found that this supplement

improved the quality of sleep and also cognitive performance during the day. There are more data on magnesium bisglycinate than threonate for sleep but these seem to work interchangeably for this purpose.

• Apigenin: This compound is among the most widely studied flavonoids (chemicals found in plants) and is present in chamomile, parsley, oranges, celery, and thyme, among herbs and vegetables. A growing body of evidence supports apigenin as a sleep aid. Animal studies have found a mild sedative effect from apigenin. Studies in humans also indicate a sleep benefit. In one investigation, apigenin mildly improved daytime function among people with insomnia. Among a group of new mothers suffering from poor sleep and depression, apigenin was found to improve both.

• Theanine: Evidence suggests that theanine increases levels of GABA, an amino acid that, when broadly increased in the brain, may reduce anxiety.

• Glycine: Research suggests glycine works by elevating serotonin, which helps regulate sleep, and activating NMDA receptors in the SCN, where the circadian clock is headquartered.

• GABA: As noted earlier, GABA is an amino acid associated with reducing brain activity. It promotes relaxation and is therefore conducive to sleep.

• Melatonin: As discussed in Sleep Protocol 3, we produce natural melatonin, a hormone that causes us to feel sleepy. This production surges at night, leading to feelings of sleepiness. This hormone works well all on its own. However,

cherry juice for seven days. This preliminary evidence suggests a potential benefit, with larger and longer-term studies needed. Tart cherry juice is safe to try as a sleep aid taken shortly before bedtime. Tart cherry extract capsules are preferable if you don't want the added calories from the juice.

• Kiwi extract: A small study of twenty-five people who self-reported their sleep patterns concluded that kiwi may modestly improve sleep duration and efficiency. Consuming kiwi fruit with the skin on an hour before bed may be a mild sleep aid. More evidence is needed to be certain of this benefit, but kiwi is safe to try. Kiwi extract is a good noncaloric, capsule-based approach to get the same effect. (And you don't have to eat kiwi skins!)

## FREQUENTLY ASKED QUESTIONS

### When should I take supplements for sleep?

It's best to take any sleep aid supplements about thirty minutes before you intend to be asleep. Magnesium supplementation during the daytime can be useful for other reasons, but if you're taking it specifically for sleep, best to take it before bed. The one exception is theanine, which some people like to take when they drink caffeine, in order to offset the jitters that they experience from caffeine. In fact, that is one reason energy drink manufacturers often include theanine in energy drinks.

I take AGZ, a powdered sleep supplement that you mix with water and drink. (Note: I helped design AGZ.) It contains many of the compounds listed above. It remains the best sleep supplement I have ever taken, and yes, I am incentivized to say that, but I am measuring my sleep stats most nights. I am able to take half a packet (one full packet is the recommended dose). It has outperformed all the other sleep pill/powder combinations I have tried over the years and even the "sleep peptide" I have tried called pinealon. There are, of course, other great sleep supplements with multiple compounds out there too. Compare and contrast ingredient lists and costs, and once again: Only by trying individual single-ingredient supplements can you really determine what does or doesn't work for you.

**What brand of supplements should I use?**

Always look for third-party testing. The NSF (founded as the National Sanitation Foundation) or US Pharmacopeia (USP) offer rigorous testing for supplements. "NSF Certified for Sport" indicates that a product met the review standards of the NSF.

▸ Approach Prescription Sleep Drugs with Caution

## WHAT TO DO

• Talk to your doctor. If you're still struggling with sleep after working through the prior sleep protocols, and you have tried the stress-management protocols, too, talk to your doctor. There may be an underlying health condition that needs addressing. Or it may be that your sleep issues warrant more intensive treatment. Cognitive behavioral therapy has shown efficacy for sustained sleep disturbance.

• Stay away from prescription sleep aids if you can. These are best avoided, unless your physician prescribes them as a last resort. Most can carry side effects, like confusion and daytime drowsiness. Some, like Ambien, may harm sleep architecture and glymphatic function. That means even if you have more time in bed, sleep might not be doing the "cleaning" job it's designed to do. Some, like trazodone, and the DORAs

(dual orexin receptor antagonists), appear safer than the others, yet it is easy to become dependent on them for sleep once

## HOW IT WORKS

of Sleep Medicine has recommended the use of certain antidepressants as a treatment for insomnia among people with depression, but clinical evidence for this benefit is limited. Trazodone, a serotonin receptor agonist and reuptake inhibitor (SARI) seems to preserve sleep architecture better than other antidepressants when taken specifically for sleep. Please note that many people who take selective serotonin reuptake inhibitors (SSRIs) like Prozac, SARIs like trazodone, or norepinephrine-dopamine reuptake inhibitors (NDRIs) like Wellbutrin/bupropion during the daytime for depression, anxiety, or other reasons do not experience altered sleep, and some of these, like bupropion, when taken early in the daytime, can indirectly improve sleep later that night. Psychiatrists now will prescribe antidepressants specifically for sleep because some can improve sleep.

- Melatonin receptor agonists: This category of medications mimics the action of melatonin and may help with falling asleep faster and improving sleep quality. They do not appear to cause dependency or serious side effects. However, the efficacy of melatonin receptor agonists is somewhat unclear.
- Anticonvulsants and dopamine agonist drugs: These may be prescribed to treat restless legs syndrome and other conditions involving nighttime limb movement that can make it difficult to fall and stay asleep, as well as to treat pain. Examples include gabapentin and cabergoline.
- Dual orexin receptor antagonists (DORAs): Insomnia is characterized by abnormal wakefulness, and orexin, a neuropeptide that helps regulate the sleep-wake cycle and that is altered in narcolepsy, has been found to cause a state of arousal. Thus it makes sense to investigate orexin as a potential target for insomnia drugs. That led to the development of the DORAs, which work by reducing the action of orexin and thus wakefulness circuits in the brain rather than by direct sedation. Early evidence of such drugs shows that they do effectively block orexin, and indicate a potential role for them in safely treating insomnia. A meta-analysis of ten randomized clinical trials that combined a sample of 7,086 individuals with insomnia found these drugs to be generally safe, with a comparably lower incidence of mild side effects and with improved sleep compared to a placebo. These new

medications are expensive, typically costing about $300 or more per month. Generic lower-cost versions are not yet available.

▸ Adjust Your Caffeine Schedule to Improve Your Sleep

## WHAT TO DO

• Avoid caffeine in the eight to ten hours before bed. As it's a stimulant with a long half-life, caffeine should be avoided late in the day, especially if you have trouble falling asleep. Some people who are especially caffeine sensitive may have to avoid caffeine in the twelve hours before bedtime. Even those who can "fall asleep after drinking coffee" are likely not getting the best sleep they could. As someone who loves caffeine, I relate to anyone reluctant to stop drinking caffeine in the afternoon. As painful as it can be, I personally avoid it after 2:00 p.m. most days.

• If you tend to experience an energy crash in the afternoon despite sleeping well, consider delaying drinking caffeine until thirty to ninety minutes after waking up. This slight adjustment to your caffeine schedule may help you have more sustained energy levels throughout the day and may help you avoid the need to drink caffeine after 2:00 p.m.

## HOW IT WORKS

improved their capacity in all these domains. The findings were favorable toward Coca-Cola because the chemical in question seemed to only help people, not hurt them, at least in the context of what was measured. But Coca-Cola wasn't interested in sleep. (If you were wondering, the US government's case was dismissed on a technicality when the company argued that caffeine wasn't added to the beverage but rather was naturally occurring.)

According to Hollingworth's report, published in the American Journal of Psychology in 1912, a few grains of caffeine taken on successive days were enough to impair sleep. The connection relied on self-reporting and each individual recording their impression about the quality of their sleep following each study day. But the findings were consistent. Decades later we know how that happens: Caffeine leads to longer sleep latency (that is, the time it takes to fall asleep) and more nighttime waking, thus leading to reduced wakefulness the next day. We also now know that caffeine can alter sleep architecture, including less time spent in REM.

Hollingworth's study was the first of many to come. Today we understand a great deal about how caffeine stimulates brain activity. It's a psychoactive stimulant that increases the actions of epinephrine and dopamine, in part by increasing the availability of dopamine receptors, the sites on cells where it can attach and thereby initiate its effects. Caffeine also increases the activity of certain neural circuits and brain areas associated with alertness, such as the locus coeruleus, which we discussed earlier in this chapter.

We also understand now how it interferes with sleep. That brings us back to adenosine. Caffeine is what is called an adenosine antagonist. It parks in adenosine receptors, which prevents adenosine from exerting its normal fatigue-inducing activity in the brain. As sleep expert Matthew Walker describes it, "Caffeine keeps pulling out the chairs from under adenosine." So even though adenosine is building up with each hour we are awake and increasingly so when we are very alert or active, our brains don't register that fatigue or the number of hours we've been awake accurately. Caffeine dampens the communication between adenosine and the brain. This process explains the phenomenon many of us know of as the caffeine crash. Once caffeine's actions on those receptors wear off, which our metabolism accomplishes in a few hours, the non-receptor-bound adenosine that has accumulated suddenly can bind and fatigue hits us. In addition, caffeine makes our brain more active and increases our capacity for physical effort, and that generates more adenosine as a byproduct, which in turn means the crash is harder than it would be without caffeine.

This all sounds rough, but I want to reiterate: I love caffeine and I have no plans to give it up. The data say caffeine in moderation is fine... and may even be good for us. A recent large-scale study involving over 100,000 subjects found that daily intake of approximately two to three cups of caffeinated coffee or tea can reduce risks of stroke and dementia.

a person who is in caffeine withdrawal is back to a baseline state of sleep. And, as I've mentioned, people's sensitivity to caffeine varies. That being said, the evidence shows that our sleep patterns tend to adapt to a daily dose of caffeine. Regular caffeine consumers are able to fall asleep, stay asleep, and cycle through the stages of sleep. It appears that chronic caffeine users develop a sort of sleep-tolerance for the stimulant. This is good and bad. Good, because it means folks like me can enjoy a morning yerba maté and a 2:00 p.m. espresso and still fall asleep easily at 10:00 p.m. Bad, because there is a chance that anyone drinking caffeine past noon is getting less than optimal sleep. Not everything in life is about getting the best possible sleep of course, but if you are struggling to feel

rested enough in the daytime or are waking after seven to eight hours of sleep and still not feeling alert, you may try avoiding caffeine in the eight to ten or even twelve hours prior to sleep. Note how you feel. Those tracking their sleep with Eight Sleep, WHOOP, Oura rings, or other devices can see how this impacts their REM and non-REM sleep.

The recommendation to delay caffeine by sixty to ninety minutes after waking is just a suggestion. Indeed, this suggestion—when I first made it on the internet—made a lot of people VERY upset. (People don't like it when you try to take away their morning coffee). To be clear: You don't have to do it. It is also not supported by a clinical trial. It is, however, supported by ample mechanistic evidence of how caffeine adjusts adenosine, and how adenosine continually reduces in the first one to two hours of the day if we don't ingest caffeine. If you are someone who tends to experience energy crashes in the afternoon and you rely on caffeine to overcome them, I suggest at least trying to delay your morning caffeine by sixty to ninety minutes for a few days. It can be a bit painful at first, so maybe start with a thirty-minute delay, then sixty minutes, then ninety minutes, and see if you still need the afternoon caffeine kick. Then try NSDR, as described in Sleep Protocol 8, if you're having trouble making it through the afternoon without it.

One key thing: If you like to exercise soon after you wake up and need caffeine to do so—as I and many others do—then by all means, drink it upon waking and do your workout. Just be sure to hydrate well too. Caffeine can be a mild diuretic. It's much less of a diuretic than once thought but we tend to wake up slightly dehydrated as it is. Before you slam back two coffees, it's probably best to hydrate first thing.

### In My Experience

For my first caffeine of the day, I usually drink yerba maté, a Paraguayan (and Uruguayan and Argentine) tea that is traditionally sipped from a small gourd through a metal straw but that can also be brewed and consumed like any other tea. I'll drink it hot or a cold-brewed canned version. Mateina is my favorite. (Note: I am a partial owner of Mateina, but I love it without bias. Also, it is a good idea to avoid drinking smoked varieties of yerba. Mateina is air-dried, not smoked.) I find that the alertness effect from yerba maté is more gradual and causes fewer jitters, and results in a milder, if any, "crash" as it wears off. Sometimes I have coffee or espresso later in the morning. My absolute favorite time for coffee is in the early afternoon, after a twenty-to-thirty-minute NSDR session. Talk about a reset. My best work sessions seem to follow this protocol. However, my

sleep suffers if I drink coffee after about 3:00 p.m. I can fall asleep perfectly fine, but my sleep is not as deep (I use a sleep tracker, and it confirms that I get much less REM sleep after late-day caffeine). So I try to limit coffee and other sources of caffeine after 3:00 p.m. That's not a hard-and-fast rule; sometimes I drink coffee late in the day, but that's rare and I always regret it when I'm still awake at midnight.

I like to exercise first thing in the morning, within an hour or two of waking up, so on workout days my target regimen is to drink water and caffeine soon after waking, view sunlight, and then exercise. On days when I cannot exercise early, I tend to delay caffeine intake until midmorning, which I find keeps my energy levels steadier into and throughout the afternoon. On these days, I often skip afternoon caffeine because I don't need it.

## FREQUENTLY ASKED QUESTIONS

### Does the source of caffeine matter?

No. Different drinks contain different amounts of caffeine, and chocolate has small amounts too. But the caffeine molecule works pretty much the same regardless of where it comes from. Some drinks like yerba maté and guayusa tea also contain non-caffeine stimulants in them like theobromine—a plant alkaloid also with antioxidant properties. And of course nowadays there are hundreds of different energy drinks available that combine caffeine, neuromodulator precursors like L-tyrosine (the amino acid from which dopamine is made), taurine, and on and on. I will drink a zero-sugar energy drink now and again. I discuss these and other stimulants more in chapter 6.

### Is caffeine unhealthy?

Studies have consistently shown a health benefit from coffee, particularly when consumed before noon, up to a certain amount. Generally, beyond four cups per day of coffee, or the equivalent of 400 milligrams, there are diminishing returns or possibly negative consequences, although larger individuals may be able to handle more than that with no problem. Many of the benefits from coffee and tea are not from caffeine but from antioxidants contained in coffee beans and certain teas. That finding is supported by the fact that decaffeinated coffee confers the same benefit as caffeinated coffee, which includes a potential reduction in the risk of heart disease, prostate cancer, melanoma, endometrial cancer, liver conditions, type 2 diabetes, and neurodegenerative diseases. In the case of yerba maté, research suggests that it may facilitate weight loss by reducing appetite and perhaps by its effects on digestion as well as the microbiome, a critical aspect of health we will address in chapter 4.

### Avoid Alcohol Within Eight Hours Before Sleep

**WHAT NOT TO DO**

- Alcohol should not be used as a sleep aid. Alcohol severely disrupts the elegantly sequential architecture of sleep, and reduces both the total amount of deep sleep and REM sleep, so even if it helps you drift off and you sleep for seven to eight hours, it will negatively affect the quality of your sleep.
- If you drink, avoid alcohol within eight hours of the time you plan to be asleep. This timing reduces the amount of alcohol in your body by the time you go to bed.

The "What Alcohol Does to Your Body, Brain & Health" episode of the Huberman Lab podcast remains one of the top downloaded and shared podcast episodes of all time, across all podcasts. It makes clear that the data show zero to two drinks per week is fine for nonalcoholic adults without confounding medical issues. However, zero is better than any, and there is growing data on the

## HOW IT WORKS

Alcohol has also been found to decrease the amount of growth hormone and testosterone released during the night—hormones critical for vitality and metabolism for both men and women. Normally, growth hormone is released relatively soon after falling asleep, and testosterone is released before and during REM sleep. The negative effects of alcohol on REM and these hormones happen acutely, meaning after even just a single night of drinking or for those who are alcohol sensitive, even one drink. Only if we metabolize the alcohol prior to bedtime do we avoid some of these consequences, which is why this protocol recommends a specific timing.

Chronic drinking is even more problematic for sleep. Insomnia and alcohol dependency frequently coexist, for example. The cycle is self-perpetuating in that poor sleep leads to increased caffeine and other stimulant use during the day and into the afternoon and evening, which leads to difficulty sleeping at night, which leads to alcohol intake as a sleep aid. Beyond the changes one or two drinks can make on REM sleep, people with alcohol dependency tend to have decreased amounts of both N2 and N3 sleep, the non-REM period when we sink into a deep sleep. These alterations in sleep architecture can persist for some time even after a person stops drinking. Frequent drinking also interferes with our circadian rhythm, disrupting or even decreasing the amount of melatonin our brains produce.

Adolescents are at particular risk of turning to alcohol intake as a sleep aid. In 2008, public health researchers at the University of Texas Health Science Center reported their findings from a large, yearlong study of insomnia among youth between the ages of eleven and seventeen years. They collected data on 3,134 people in this age range from a baseline date and then twelve months later and found that chronic insomnia heightened the risk of alcohol use. That's concerning because insomnia is the most common sleep disorder among young people, and because nighttime phone use, which is very common in this age group, also interferes with sleep quality and duration. Phone use at night has been linked to insomnia. That means there are two major risks to this age group when it comes to good sleep, not to mention the chronic stress that kids at that age face. We'll discuss phone use in Sleep Protocol 14.

**FREQUENTLY ASKED QUESTIONS**

**It's not practical to avoid alcohol within eight hours of sleep because any drinking I do happens at night. How can I compensate for the negative effects of consuming alcohol close to bedtime?**

For most people, social drinking happens well within eight hours of going to sleep. What's important here is that drinking this way every night, multiple times per week, or even once per week will disrupt sleep, and thereby health, in detrimental ways. Occasional alcohol use closer to bedtime, say once a month or less, is very unlikely to cause long-term problems, unless it's excessive during that once-a-month night out. If you drink in the eight hours before sleep, make sure you also consume sufficient water before going to sleep and after waking. Take electrolytes too if you can, and emphasize other behaviors that support great sleep, as well as a healthy microbiome, to try to offset the effects of alcohol.

To be clear, it's good to enjoy life. If having a few drinks during a celebration or a night out now and again is how you do, you're not alone. The important considerations are that this isn't happening too often, that you're not driving, that you're not experiencing any kind of dependency, and that you're not too young to drink (the negative effects of alcohol are much worse for the developing brain compared to the adult brain).

**Does the type of alcohol matter?**

Not with respect to the negative effects on sleep and the microbiome. As mentioned above, no alcohol is better than any alcohol. Although many studies have been done on the potential benefits of red wine, conclusive evidence of this link does not exist. The amount of red wine you would need to drink to get benefits from grape skin, for example, would require

consumption of enough alcohol to far outweigh any benefits. Put simply, the interaction between alcohol and our sleep processes is the same regardless of the drink.

▸ Avoid Cannabis as a Sleep Aid

## WHAT TO DO

• Do not use cannabis specifically for sleep. Cannabis should not be used to facilitate falling asleep or returning to sleep after night wakings. It potently reduces REM sleep.

• Understand CBD's impact on sleep. Some evidence exists on the benefit of CBD for sleep but those effects are mainly due to the reduced anxiety it causes. At higher dosages (e.g., 160 mg or more), CBD is more of a sedative. The dosage of CBD that makes it an effective sleep aid varies depending on weight and other factors.

## HOW IT WORKS

Cannabis is a psychoactive drug that comes from the cannabis plant. The main chemical behind its psychoactive effects is THC (delta-9-tetrahydrocannabinol). CBD, or cannabidiol, is another well-known component of the cannabis plant; this chemical technically is psychoactive, too, because it impacts the brain, but it's not responsible for the "high" that people are usually seeking when they use cannabis. THC can impair cognitive or behavioral functioning; CBD doesn't

the fact that examining this connection was not the main objective of the investigations somewhat limits the usefulness of the findings. Furthermore, THC and CBD are vastly different, which makes any reference to cannabinoids confusing and unhelpful.

A survey of 4,637 ninth through twelfth graders in Colorado found that those who consumed cannabis through edibles or vaping tended to sleep fewer than seven hours per night. Another study of adolescents found that the sedative effects of THC were followed by increased wakefulness at night and impaired memory the next day, which could have been directly due to the THC, which is known to impair memory, lack of sleep, or both.

Researchers have also found a potential link between use of cannabis among pregnant mothers and poor sleep habits among their children during adolescence. You might be asking, "How many pregnant women use cannabis?" A recent report said that anywhere from 4 to 12 percent of

pregnant mothers in the US said they'd used cannabis during pregnancy, usually by edible and usually in the first trimester prior to learning they were pregnant. Cannabis use during pregnancy has reportedly increased in recent years, from 3.4 percent in 2002 to 7 percent in 2017. Smoking or ingesting cannabis during pregnancy has been tied to lower birth weight, possibly due to THC-induced changes to the placenta. Rates of neonatal and infant death are higher with cannabis use during pregnancy compared to no such use (and combining cannabis with nicotine raises this risk even higher).

Many people use CBD to try to induce sleep, but rigorous evidence of a direct benefit to sleep is currently insufficient. Chances are any positive effects of CBD on sleep are due to its anxiety-lowering effects. CBD may help people to stop ruminating—a prerequisite to falling asleep. As such, there are populations for whom CBD may be useful. In one study, fifty-seven of seventy-two adults with anxiety reported improved sleep with the use of CBD, although this study was retrospective, meaning the participants were reporting their recall of possible effects from memory rather than measuring them directly. A 2025 review of twenty-nine studies identified a potential for CBD to reduce sleep disruptions but emphasized that research on this link has suffered from a lack of standardization. There certainly is enough evidence to warrant continued study of CBD in this context and no known negative effects of CBD that would preclude trying it for sleep if you feel stressed at night.

That said, in January 2024, a team from Australia published the results of a small, randomized, placebo-controlled study in which fifteen people received 150 milligrams of CBD and fifteen people received a placebo as a sleep aid to combat insomnia. The investigators observed no difference between the two groups in the time it took to fall asleep, the efficiency of their sleep, the duration of sleep, or overall severity of insomnia. There were some signs of an improved mood upon waking among the CBD group, and that group had fewer night wakings after two weeks of treatment.

One more thing: An increasing number of studies report a potential for THC (but not CBD) to induce psychosis in certain individuals. If your goal is to improve your sleep, there are better tools covered in the other protocols of this chapter. The same goes for stress management—more reliable tools exist (and these can improve sleep too), which I'll explain in a later chapter.

▸ Avoid Screens Within an Hour or More Before Bedtime

## WHAT TO DO

• Avoid screens in the bedroom at night. This includes phones, laptops, tablets, and TV screens.

• Avoid consuming information on an electronic device within an hour (at least) of bedtime. Light from phones and other screens can disrupt your circadian clock.

• If you opt to read or watch content on a screen in the hour before sleep, dim the screen. I suggest lowering the brightness and/or activating the "red" screen function. You can also wear blue and green (short wavelength) blocking glasses.

## HOW IT WORKS

In 2011, the National Sleep Foundation conducted a poll. They asked 1,508 Americans about their sleep habits, whether any technology was around them in the hour before bed, and if so, what it was and how they used it. Ninety percent of respondents said they used some kind of screen device at this time; for those under thirty, that percentage climbed to 96! Adolescent

researchers at Harvard Medical School reported that people who read on an e-book at night with a dim bedside light took longer to fall asleep, were less sleepy, had lower levels of melatonin, and were less alert the next morning compared to people who read a printed book with a dim bedside light before bed.

And there's more. In 2018, a team of researchers, including UCLA psychiatrist Peter Whybrow, reviewed the data on the potential link between LED lighting and mental health. They concluded that exposure to blue light, in particular, posed a risk to sleep and other processes governed by the circadian system. There is also evidence suggesting that the way in which artificial lighting at night interferes with this system may lead to depression.

The problem with devices at bedtime is not solely due to the type of lighting. Technology, particularly interactive technology, stimulates our brains in a way that counters the calm that we need in order to fall asleep. For example, the devices in the 2018 study were most strongly linked to difficulty falling asleep and to poor sleep quality. Also, people who sleep with their phones in their rooms with the ringer on contributed to the numbers reported in the study, indicating that the problem isn't just about late-night scrolling.

We have known since at least the 1970s that the bedroom should be reserved for sleep and intimacy. The evidence linking various stimuli to insomnia is well established. Yet decades later, here we are, with about 1,350 out of 1,508 randomly selected Americans ignoring this basic practice.

Believe it or not (believe it!), this protocol is one of the most potent in this chapter. It's also one of the most difficult to stick to for many people. Sleep-based apps and the use of smartphones as alarm clocks don't make it any easier. Still: I highly suggest you try it.

**FREQUENTLY ASKED QUESTIONS**

**Do blue-blocking glasses counter the negative effects of screens at night?**

Yes, somewhat. Better, however, would be glasses that block both blue and green light, aka short wavelengths of visible light, at that time—especially if you use screens at night. (See Sleep Protocol 3 for more information.)

## A Tip Sheet for Avoiding Jet Lag

Some people experience terrible jet lag, others far less so. This changes as we age. And it depends on where we fly—which direction and if we need to push ourselves to wake earlier or stay awake later than usual. The latter is always easier for people.

Five factors allow you to shift your circadian clock. Knowing these can help you switch to a new time zone quickly, which allows you to be more effective on work trips and enjoy vacations more fully.

The factors you use to entrain your SCN to a new local light-dark cycle are, in descending order of influence:

- Light
- Temperature
- Meal times
- Exercise
- Social factors

You might think: Can't I just hop on the local schedule? Alas, that's hard because your SCN is entrained to wherever you have been for the prior week or so before your trip. So for instance if you fly from California to London and arrive at 5:00

by using light, exercise, and caffeine. In this case, however, you'll want to shift everything thirty to forty-five minutes later in the day as opposed to earlier. A slightly later wake-up time, a slightly later caffeine intake in the morning and early afternoon (but still avoid caffeine in the eight to ten hours prior to bedtime), and a slightly later workout—even just by forty-five minutes—will all help. You'll also want to make sure that you view bright light from artificial lights one to two hours after sunset. Remember, you're trying to prepare to stay up later as you soon fly westward.

### WHEN YOU ARRIVE AT YOUR NEW DESTINATION (REGARDLESS OF WHETHER YOU FLEW EAST OR WEST)

- Eat like a local. Adopting the local meal schedule when you arrive—eating breakfast, lunch, and dinner more or less at the same times as the locals, even if you're not hungry—will help you shift your circadian clock to the new time zone. Personally, I hate to eat when I'm not hungry (and love to eat when I am!), so I opt out of this protocol. If it works for you, go for it. The sooner you can get on the local meal schedule, the better. Many find it effective to fast for a period of fourteen to sixteen hours before the local breakfast time so they are very hungry when that time arrives.
- Time your light exposure correctly by knowing your temperature minimum. This is the real power tool to shift your circadian clock fast, but it involves learning your temperature minimum first. See Sleep Protocol 3 and Sleep Protocol 5 for more on that topic.

For now, know that it occurs approximately two hours before your usual wake-up time. If you typically wake up at 7:00 a.m., your temperature minimum is very likely to be between 4:30 a.m. and 5:30 a.m. Here's the key: If you get bright light exposure in the one to two hours after your temperature minimum, you will adjust your circadian clock such that it will make you want to go to sleep earlier on the subsequent nights and wake up earlier on the subsequent mornings. Perfect for traveling east. Conversely, if you get bright light exposure in the one to two hours before your temperature minimum, you will shift your circadian clock such that it will make you want to go to sleep later on the subsequent nights and wake up later on the subsequent mornings. Perfect for traveling west. But remember we are referring to the time of your temperature minimum before you leave! This is vital to understand. Get it right and you'll shift fast across time zones. Get it wrong and it can be a mess. Why? Well, think about this. Imagine you fly from San Francisco to London (+8 hours). You land in the morning London time and the sun is out. I said get morning sunlight in your eyes, right? Yes, but for the first two to three days in

London before you adjust your clock, viewing that London morning sunlight is the same to your brain's clock as getting a giant blast of light in your eyes at midnight San Francisco time. So you just made your body want to go to bed much later and wake up later. You got it backward.

Many people sleep well their first night after flying east, wake up, see sunlight, and the second night is a mess. They can't sleep! It would be better to wear dark sunglasses during the two to three hours before your temperature minimum for the first two days in London. After those first two or three days, just view sunlight per the recommendations earlier in this chapter. Obviously, I am just using London as an example. You must figure it out based on your own travel destination. I know it sounds complicated, but if you know the time of your temperature minimum back home, and follow the instructions above, you'll nail this protocol easily.

- Use artificial light and blue and green (short wavelength) blocking glasses as tools. Of course, it's often very cloudy, or you may land at night, and we can't control the time of sunrise and sunset—though if you can, please contact me. What you can do, however, is travel with a portable bright light device (you'll want one with 5,000- to 10,000-lux brightness) to shift your circadian clock. Several are the size of a book and thin enough to fit in a carry-on bag. I have no affiliation with any companies that make these, but I use the Verilux Happy Light when I travel. They cost about $50. At home I use a larger countertop unit, the Carex Daylight (also no affiliation) and Ra Optics or Bon Charge full-spectrum LED bulbs in the daytime, as discussed earlier.

Remember that even when it is cloudy, sunlight through cloud cover is still almost always far brighter than artificial lighting. Get outside and view that sunlight at the correct times and you'll shift your circadian clock fast. Then, you can return to the standard protocols to stay on the local schedule.

One more thing we can all control is the brightness of our evening and nighttime environment. As you get ready for bed, reduce exposure to bright lights—especially blue and green, or short-wavelength, light. Alas, as you now know, all LED lights are rich in blue and green wavelengths. For that reason, I wear blue and green (short wavelength) blocking glasses after sundown to help adjust to the local schedule. This protocol is known to protect against the melatonin-reducing effects of bright light and to prevent the cortisol-increasing effects of bright light.

- Maintain a regular exercise schedule. While traveling, try to maintain a consistent early-in-the-day exercise routine (adjusted to the local

schedule). That, too, helps shift your circadian clock.

- A final tip. For trips that last fewer than two days, staying on (or near) your home sleep-wake, eating, and exercise schedule can be beneficial to minimize disruptions to your circadian rhythm.

These tips will help you quickly overcome any jet lag in your future travels—they've certainly helped me over the years, as well as people in the military, medical professions working night shifts, and more.